CHAPTER 12A

Supply Chain Management (SCM)

12.1 DEFINITION OF SUPPLY CHAIN

Supply chain is the entire process of accepting a customer order through the delivery of the product to the customer inclusive of supply procurement and production of the product. A supply chain is a collection of interdependent steps, when thoroughly followed gives raise to a certain objective as meeting customer requirements. SCM is a generic term, encompassing:

- The coordination of order generation
- Order taking
- Offer fulfilment/distribution of products, services or information.

Numerous, independent firms and customers are involved in a supply chain such as manufacturers, component/part suppliers, parcel shippers, senders, receivers, wholesalers, retailers, etc.

Supply chain is simply the combination of tasks wherein any company would like to perform to move services or products from suppliers to customers. Though the definition appears to be simple, it is complex when the question of building a model of a generic supply chain can be commonly used by all companies, e.g., Software selling company with a tractor manufacture. The difficulty in defining a generic supply chain model is that:

(i) There are many variations of supply chains almost equivalent to the number of companies.

(ii) Each company's participation in the supply chain varies in degrees.

(iii) We have different categories like physical supply chains and vertical supply chains whereas in the former type, physical product moves through an organisation and in the latter case, no physical product moves (no inventory, but website connect, buyers and sellers).

12.2 DIFFERENT CATEGORIES OF SUPPLY CHAIN

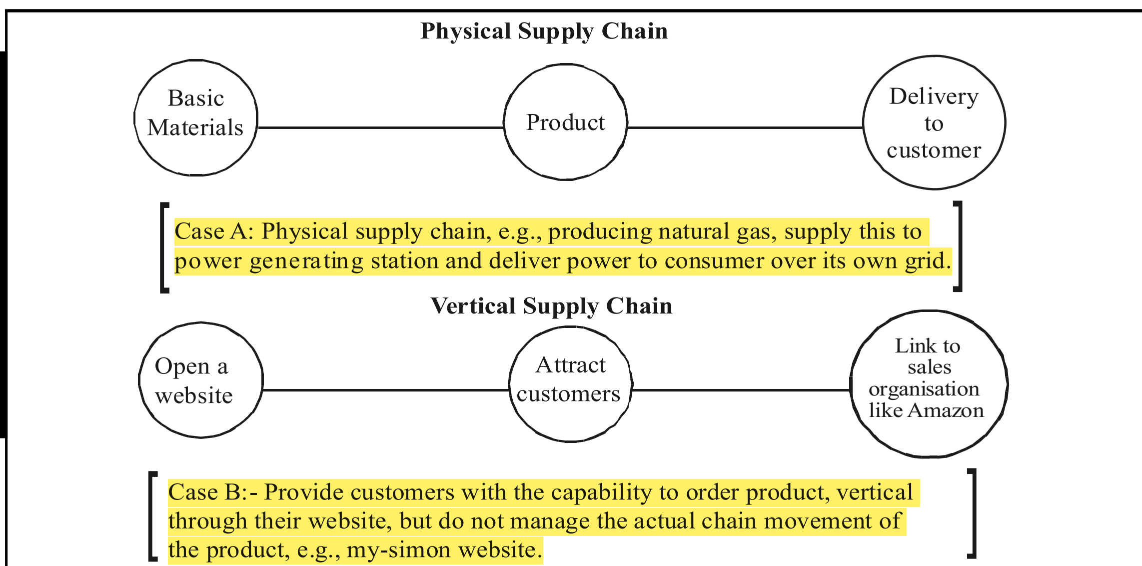

Fig. 12.1 – Different Categories of Supply Chains

Earlier, EDI was extensively used in linking and managing the entities into a virtual organisation. It had promised such revolutionary gains but not providing benefits to small and medium companies. Presently, the WWW and extranets which is also known as connected intranets have shown higher potential in this area. The splendid nature of e-commerce is its power in tackling all types and sizes of firms to interact.

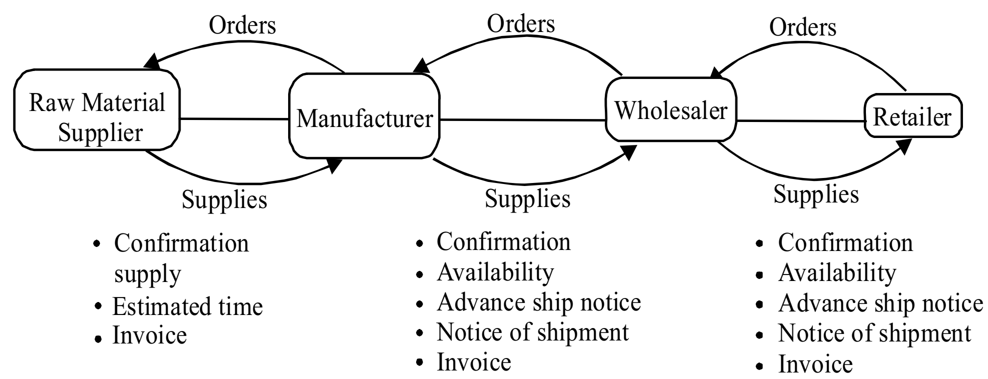

Fig. 12.2 – Information Flow in Business-to-business (B2B) Supply Chain

12.3 DEFINITION OF SUPPLY CHAIN MANAGEMENT (SCM)

Supply chain management are integrating management practices and information technology to optimise information and production flows among the processes and business partners within a supply chain. SCM is a management concept that integrates the management of supply chain process.

In the earlier years, companies used to compete neck to neck with each other. As the net age approached, situations are changing and the company with the best supply chain is going to win in the competition battle. In view of this, most of the manufacturing firms are going globally for outsourcing the best quality components. It is therefore evident that supply chain management (SCM) focuses more on those tasks that add real value to the product and at the same time results in maximum profits to the firm by leaving everything on the shoulders of the suppliers. Examples are: (i) Maruti offers better seating systems say bucket seats – it is not from their company but such seats are procured from their vendors, basically the vendors are the designers and creators of such seats. This is basically supply chain management. (ii) So, also, we find Reebok Shoes Company outsources its entire product ranges from its suppliers across the globe and Bata in India also outsource to vendors for some of the products. Reebok limits itself to the role of mere marketing. In fact, the big companies would force the vendors to supply better designed products to them, which they themselves would not have designed such products supplied by the vendors, if they had taken the design themselves. In other words, effort and concentration would have lacked, mainly due to their size and unwieldiness.

Supply chain management is a generic term that encompasses the coordination of order generation, order taking, and order fulfilment of distribution of products, services, or information.

The passage explains that large companies often get better product designs from vendors than they could create themselves, because their own size and complexity reduce their ability to focus, innovate, and put in concentrated effort. Vendors, being more specialized and flexible, perform better in design work.

12.4 SUPPLY CHAIN MANAGEMENT SYSTEM (SCM SYSTEM)

Activities

- Raw materials
- Suppliers
- Manufacturing
- Distribution
- Retail Location
- Customer

Achievements

- To maximise customer value
- To achieve a sustainable competition

Advantages

- Flow of goods/services from raw material to final product to the customer.

For, global supply chain management systems subject to be covered are:

- Market Research Report
- Industry Analysis Report
- Regional Outlook
- Application Development Potential
- Competitive Market Share
- Forecast 2023

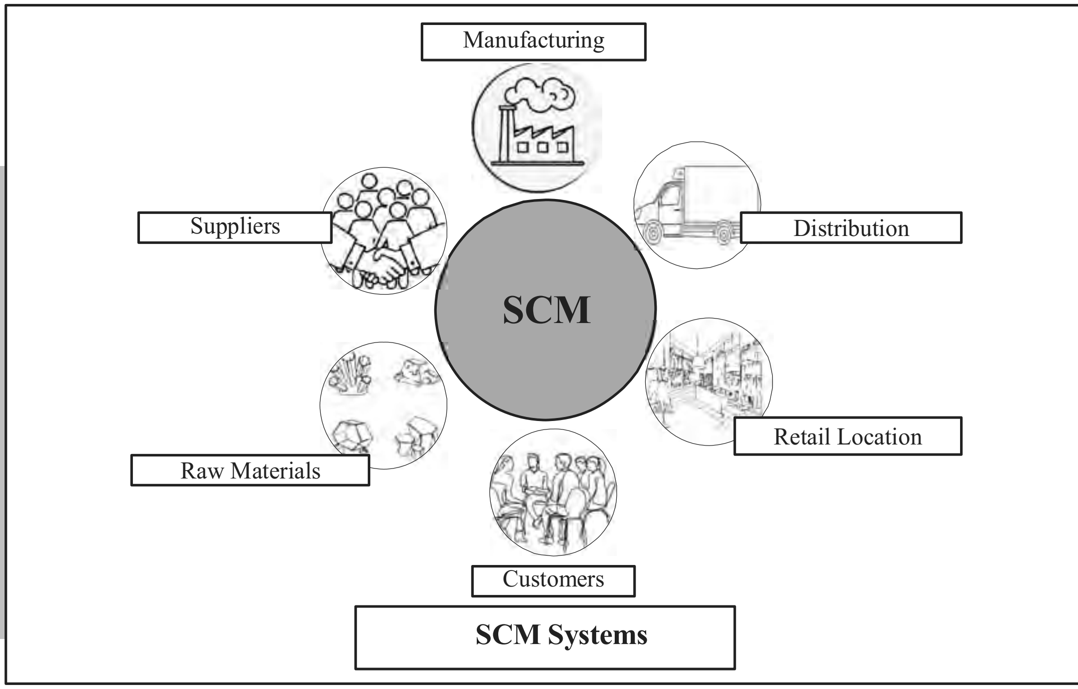

Fig. 12.3 – Supply Chain Management System

Some of the features of SCM like benefits, types, importance, significance, etc. are given below along with the definition of SCM.

Benefits

- Improvement of delivery dependability and customer orientation
- Reduction of stocks and quality improvement
- Cost reduction within the procurement, production and distribution work
- Decrease of processing time
- Avoidance of the bullwhip effect
- Strengthen relations with suppliers

The bullwhip effect is a supply chain phenomenon where small changes in customer demand cause increasingly larger fluctuations in orders and inventory upstream.

Types

- Integrated SCM and e-business suites
- Functional extended ERP systems
- Chain execution suites/software
- Specialised SCM suites
- Niche vendors

Supply Chain Management is the management of the flow of goods and services, and includes all processes that transform raw materials into final products. It involves the active streaming of a business's supply activities to maximise customer value and gain a competitive advantage in the marketplace and plays an important role with the effective management of a supply chain.

Importance

- to decrease the cost of purchasing and production supply chain management allows a business.
- to increase customer satisfaction due to its strategy.

Significance

- Important part of every organisation – small or large.
- To maximise customer value and competitive advantage.

Key goals

- Achieve efficient fulfilment of demand
- Enhance organisation's responsiveness
- Facilitate financial success
- Drive outstanding customer value
- Build network resilience

Objectives

- Creating net value
- Leveraging worldwide strategies
- Synchronising supply chain demand
- Building a competitive infrastructure
- Measuring performance globally

Importance to Growing E-commerce

- As production takes more on a global field, confirmation and control of the flow of a company's goods becomes all the more integral to ensuring smooth and productive operations.

Role

- The management of the flow of goods and services involves the movement and storage of raw materials, of work-in-process inventory and of finished goods from point of origin to point of consumption.

Top supply chain companies in 2018 are:

- Novo Nordisk
- Inditex
- H&M
- Nestle
- **Johnson & Johnson (Rank 3)**
- Intel
- Unilever
- Gartner
- BASF
- Diagro
- L'Oreal
- **Schneider Electric (Rank 4)**
- Adidas
- Starbucks
- Nike
- Magneto
- **Colgate Palmolive (Rank 2)**
- **Cisco Systems (Rank 1)**
- Pepsico
- HP Inc.
- Walmart
- Samsung
- Home Depot

Ranking in 2021

How SCM Helps in E-commerce?

The business can be through a flexible platform by providing reliability, customisation and expert support (24/7) (e.g., Magneto).

- Transforming customer's experiences
- Get to market quickly
- Reducing business costs
- Drive new revenue growth

B2B buyers expect the seamless, end-to-end ordering experience they get as consumers using e-commerce sites like Amazon. That is coresist commerce delivers Integrating Magnets and SAP delivers buyer relevant data in real time. Real time data in magnets for account level personalisation like contract/quality, pricing, inventory availability and more. Accuracy can make or break the sale here.

Magneto SAP Version

Magneto SAP ECC integration; all in one integration; SAP/R/3 integration; SAP ECC integration on HANA; SAP S/4 integration (All SAP with Magneto).

Largest Supplier of SCM Software and Software Revenue (2019)

1. SAP	$ 4122.1	4. Coup	$ 33.48
2. Oracle	$ 1734.9	5. Infor Global Solutions	$ 322.3
3. Blue Yonder	$ 835.8 (JDA Software earlier)	6. e2open	$ 168.5

Table 12.1 – Largest Suppliers of SCM Software

(As per Supply Chain Management Paper, US)

12.5 FUNCTIONS OF SCM

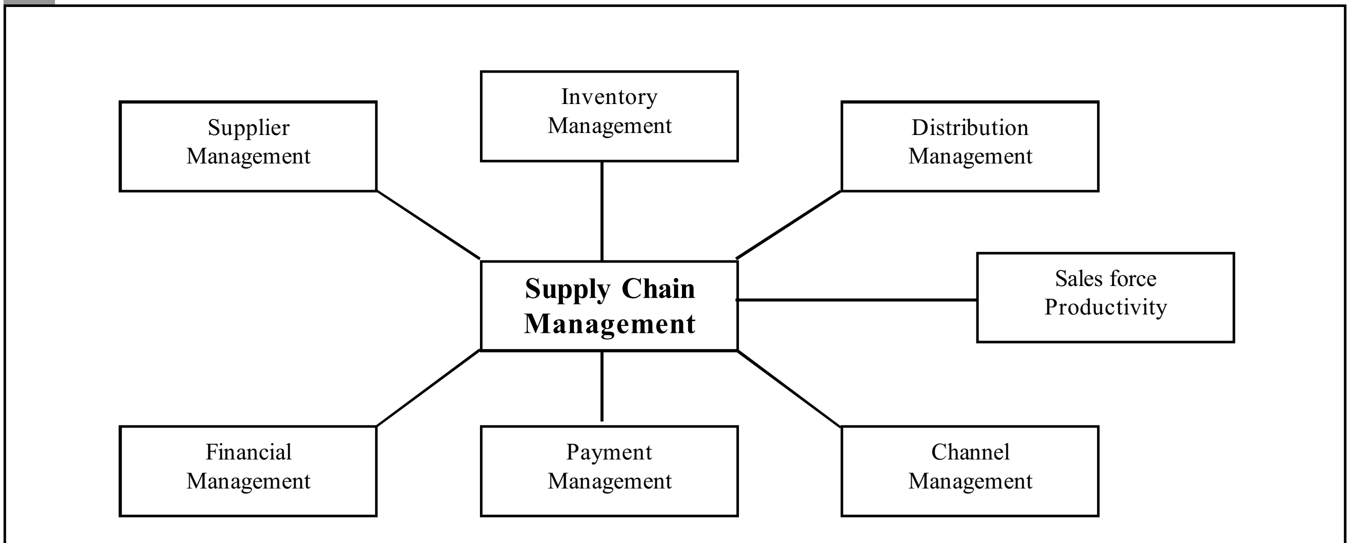

Fig. 12.4 – Supply Chain Management

(i) ***Goals of Supplier Management:***

(a) To reduce the number of suppliers.

(b) Get them to become partners in business.

(ii) Benefits of Supplier Management:

(a) Reduced purchase order (PO) processing costs.

(b) Increase number of POs processed by fewer employees.

(c) Reduced order processing cycle times.

(iii) Goal of Inventory Management: Shorten the order-ship-bill cycle.

(iv) Benefits of Inventory Management:

(a) When a majority of partners are electronically linked, information faxed or mailed in the past can now be sent immediately.

(b) Documents can be tracked to ensure they were received (audit capability improves).

(c) Reduction of inventory levels, improve inventory turns and eliminate out-of-stock occurrences.

(v) Goals of Distribution Management:

(a) To move documents related to shipping (bills of lading, POs, advanced ship notices and manifest claims).

(vi) Benefits of Distribution Management

(b) Paperwork that typically took days to cycle in the past is now fast with more accurate data allowing improved resources planning.

(vii) Goals of Channel Management: To quickly disseminate information about changing operational conditions to trading partners.

(viii) Benefits of Channel Management: Technical, product and pricing information required, repeated telephone calls and innumerable number of labour hours. Instead, electronic bulletin boards are used allowing instant access. Electronically linking production with international distributor and reseller networks eliminates thousands of labour hours per week in the process.

(ix) Goals of Payment Management: To link the company and the suppliers and distributors so that payments can be sent and received electronically.

(x) Benefits of Payment Management:

(a) Increases the speed at which companies can compute invoices.

(b) Clerical errors get reduced.

(c) Lowers transaction fees and costs.

(d) Increases the number of invoices processed (productivity)

Supply Chain Management (SCM) is considered as a philosophy for improving the enterprise's competitiveness, instead of treating it as a specific technology – and the competitive bar rises continuously. It is for optimising the delivery of goods, services and information from the supplier to the customer.

Application pertaining to Supply Chain Planning (SCP), Supply Chain Execution (SCE) known as SCM philosophies have just started to reach the surface of what enterprises can achieve with powerful computer advanced integration technologies and innovated methods of problem solving and are hoped to dramatically change soon the ways in which enterprises could be run. It is believed that enterprises will be able to respond quickly and smartly to dynamic business conditions and wish to exploit these new technologies for the advantages in SCM looking beyond ERP. The start could be by:

- Identifying the critical supply chain processes and pressure points that will improve the competitiveness.
- Assemble and integrate the portfolio of applications, along with the revised processes and measurements, which supports enterprise's objectives.

12.6 STRATEGIES OF SCM

These are four aspects worth considering in SCM.

- Strategy 1 : Prepare the firm to continuously reconfigure the supply chain on the basis of internal and external factors that influence the company. Hence, strategies are bound to vary for different companies in different industries which depends on the stage whether it is early stage or growth stage or the mature stage of the companies, where they are placed.
- Strategy 2 : Related to the customisation of the product. Here, SCM plays a very important role to deliver wide range of products catering to a large and varied type of customers at minimum cost and at the same time not sacrificing the quality of the products.
- Strategy 3 : Concentrating on the business deals between the purchasers and suppliers. In a dynamic market environment, as it is very well understood, there cannot by any long-term relationships, and it is only the competition that would decide who is the seller and who has to buy.
- Strategy 4 : SCM very much depends on the communication flow from the customer's end to the company to ensure that the specifications of the customers or the customer's ultimate needs are met by the various links in the supply chain.

In the new millennium, the supply chains are to be managed efficiently for the company's survival. The main objective of this is 'competing on the basis of your managing supply chain most effectively and efficiently'. All the other previous activities still remains. Today, the competition is very high between supply chains. The best supply chain is valued higher rather than whatever good you find in the individual packets or link of the supply chain and the overall competency if it is lacking. Several parallel forces reshaping the world of global business are contributing to this shift of the supply chain from a position where it was critical to cost and quality to time where it is becoming one of the most powerful ways for companies to offer greater and differentiated value to customers.

Things are changing fast. Several things are done as under:

(i) Manufacturing, processing of components, assembling and testing are avoided; instead straightaway assemblies are procured, tested and certified for quality by its vendor to increase efficiency and bring down the cost. Example: Blow Plast of Bombay follow this method.

(ii) Even the individual components, raw materials, and the design – that goes into the product needs specialised design and manufacturing expertise. Here, the company which procures focuses only on value-adding to inputs which would help procure good items by outsourcing from specialists, e.g., Larsen and Toubro, Hero Motors and Reebok (Reebok only do marketing). The advantage in this case is that the company receives greatest profits while leaving the rest to their suppliers.

The competitive advantage is shifting from the shop floor, where the assembling process is becoming a commoditised one to getting those inputs to the company. In other words, manufacturing to managing the supply chain.

(iii) The emergence, in the post-Net economy, of the truly global marketplace which is turning the conventional wisdom of supply chain management upside down. In the earlier days, there used to be long-term relationships with the suppliers. Your company and your suppliers were working together to improve designs, boost quality, reduce costs and share the benefits. The relationships used to be a long-term one. Now, it is totally different. It has become quicker and cheaper, and by implication, smarter to shop globally using the Net. In the Net market, every player is a global one. You are likely to discover a low-cost, high quality supplier who fits the bill. Another point worth noting is every time the supplier may be different since pricing of Net-based transactions is becoming flexible, being determined more by demand and supply at the specific time you want to make your purchase rather than by any fixed pricing strategy on the part of the seller. For example, Wipro, Bengaluru buys rubber seals from UK, Japan, US and Germany on the Net for the best bargain each time.

Hence, the supply chain can no longer be:

- Carefully cultivated
- Long-term relationship.

Hence, note the following points:

(a) Finding the right supply chain, with the strategic needs identified, will be a constant matter of choosing the right options from a menu which is itself in a constant state of flux.

(b) If a temporary equilibria is to be aimed in the dynamic supply chain market, it is not at building static, quasi-permanent chains.

The challenges seems to be tougher, as one has to abandon all the earlier practices like:

- Just-in-time suppliers
- Collaborative design.

Competition is building up in tomorrow's markets – may be a clash between competing supply chains – explaining why your supply chain is superior to other supply chains. It can be a source of competitive advantage, offers greater value to the customer than those of competitors. 80% of the value addition is done by those who supply the components and raw material, rather than the manufacturer and marketer in today's typically deconstructed value chain.

Net is changing the chain in the following ways:

- Predetermined pricing is giving way to auction-based bidding for the best price.
- Sourcing is becoming global as suppliers all over the world sell on the Net.
- Long-term partnerships with vendors are making room for deal-to-deal relationships.
- Buyers are being forced to compete with one another to secure the best and cheaper sources of suppliers.

The various strategies explained earlier are further discussed in the following paragraphs:

1. The first strategy: Here, the strategy structures the chain.

(a) The firm should be prepared to reconfigure their supply chain continuously depending on a combination of factors with internal and external to your company.

(b) The state at which your company is in its life cycle is important. This decides the supply chain activities.

The company's cycle consists of:

- Start-up
- Growth
- Consolidation/Maturity.

For example, in the TV industry, tuners and electronic gadgets undergo fast changes, and hence depends on supplier's ability to introduce new concepts and components.

The crucial relationships are as shown below:

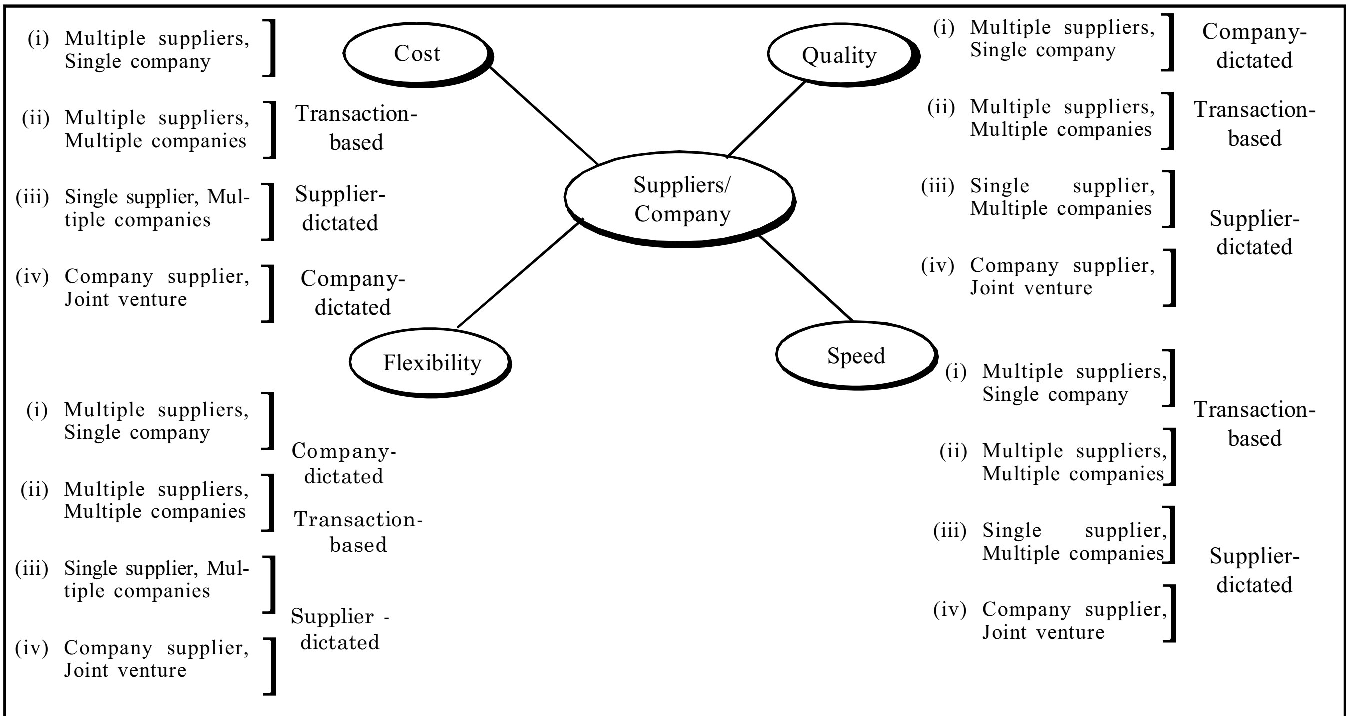

Fig. 12.5 – Critical Relationship of Suppliers/Company

Note: Basis of relationships are Cost, Quality, Speed and Flexibility.

The above will influence the nature of your supply chain, of course with more emphasis on assured availability of inputs. The focus will be on creating the chain. Finally, this will take you to established players. During the growth stage, cost will not be as much as an issue as the ability of your vendors to ramp up their supplies when the need arises – which needs a different kind of supplier from those in the first phase. During consolidation, you will be able to invest in creating suppliers and lift their technological and process capabilities and work more intimately with them in meeting customer needs. For example, Bajaj Auto helps suppliers in identifying proper layout, robotics and machinery for its vendors.

2. The second strategy: Lean is never mean.

The most successful company is the one that meets the specific needs of different customer segments. Your supply chain will have a crucial role to play provided you are being compelled to abandon the one-size-fits-all specifications for your products in favour of customisation. The supply chain should provide you with the variety you need without compromising on cost, quality or delivery. Then only you are certain to gain otherwise you will lose out to competitors who can extract such flexibility from their chains.

Quality and Strategy Changing the Chain		
Participants	**Quality**	**Strategy**
Customers	• Defining quality standards for the entire supply chain must meet.	• Customisation is the differentiator; suppliers must be chosen for their versatility.
Suppliers	• Confirming to their buyer's JIT and lean manufacturing systems.	• Customer tastes are fast changing; the supply chain has to be fast and flexible.
Manufacturers (Firms)	• Partnering their vendors in determining designs and product specifications upfront. • Passing on the costs of poor quality to their vendors instead of taking on themselves.	• In mature markets, they are building long chains and only branding the product. • Companies that compete on cost in the marketplace are picking suppliers on the basis of price.

Table 12.2 – Quality and Strategy Affecting the Chain

3. **The third strategy:** It is considered deals, not relationships.

It cannot happen to have a long relationship with vendors and suppliers cannot be taken as long-term partners, because the operations are conducted in a dynamic market environment, where competition only will decide who will sell to whom? Under such critical situations, choosing a right partner for a component or raw material who will be able to work with you in a competitive environment is needed. For a company selling products that need frequent remodelling and upgrades, it might be speed of delivery to ensure that new products can be brought to market quickly. Examples are many like Maruti Udyog, McDonald's etc. McDonald's, one of the fastest food giants, is building its own supply chain relationships. Its suppliers are, obviously, most crucial to its operations, since they provide all the inputs that go into the food it sells – on the principle of mutual need.

	Competitive Position				
Supply Chain Characteristics	**Reliable Demand Estimation**	**Unreliable Demand Estimation**	**Start-up Stage**	**Growth Stage**	**Maturity Stage**
Flexibility	Irrelevant	Critical	Critical	Important	Irrelevant
Speed	Irrelevant	Important	Critical	Critical	Irrelevant
Responsiveness	Irrelevant	Critical	Critical	Irrelevant	Irrelevant
Single-piece Flow	Irrelevant	Critical	Important	Irrelevant	Irrelevant
Cost	Critical	Important	Important	Critical	Critical
Quality	Critical	Critical	Important	Critical	Critical

Table 12.3 – The Critical Parameters

4. **The fourth strategy:**

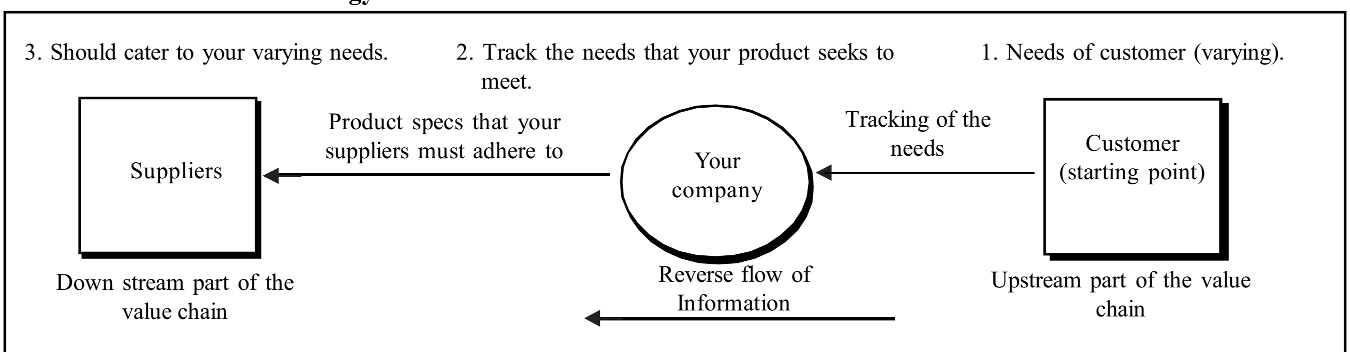

Fig. 12.6 – Fourth Strategy

Here, the steps are:

Step 1 – Information originates from the customer (Upstream end of the value chain).

Step 2 – Track the needs; decide on product changes.

Step 3 – Change in product specifications to be informed to the supplier. Supplier should cater to the varying needs of the customer.

Note: The needs of the customer goes on changing. Constant information flow should be there to the company which is crucial. Further, customer needs will be different across individuals and over different time periods.

The real solution is your supply chain must be sensitised to the exact demand pattern at the consumer end at any given point. The key to ensuring accurate information flow within the supply chain is to minimise the dissemination points of data (to filter out the unnecessary information). For example, KOEL, OTIS elevators deal with only few vendors to streamline their own data processing. Your supply chain can yield unmatched competitive advantage. But it can prove just as elusive as your customer. Companies that compete with you in the same product category may also try to steal your suppliers. Hence, the strategy for managing your supply chain will also require you to become the best possible customer for your vendors.

Cost management changing the chain:

- Demand forecast tools are being used to micro-tailor orders for supplies.
- Inventories are being demanded as companies are sourcing on a need based system.
- Manufacturers are setting target costs to their suppliers instead of asking for prices.
- Companies are helping their vendors to lower their production costs so as to pay lower prices.

12.7 SUPPLY CHAIN OPTIMISATION TECHNOLOGY

Optimisation is the technology employed within a supply chain management system that can have the single greatest impact in:

- Reducing costs
- Improving product margins
- Lowering inventories
- Increasing manufacturing throughput.

Supply Chain Management

(i)	Goals of Supplier Management	(i)	Reduced purchase order (PO) processing.
	Benefits of Supplier Management	(ii)	Increase number of POs processed by fewer employees.
		(iii)	Reduced order processing cycle times.
(ii)	Goals of Inventory Management		Shorten the order-ship-bill cycle.
	Benefits of Inventory Management	(i)	When a majority of partners are electronically linked, information faxed or mailed in the past can now be sent immediately.
		(ii)	Documents can be tracked to ensure they were received (audit capability improves).
		(iii)	Reduce inventory levels, improve inventory turns, eliminate out-of-stock occurrences.
(iii)	Goals of Distribution Management	(i)	To move documents related to shipping (bills of loading, POs, advanced ship notices and manifest claims)
	Benefits of Distribution Management	(ii)	Paperwork that typically took days to cycle in the past is now fast with more accurate data allowing improved resources planning.
(iv)	Goals of Channel Management	(i)	To quickly disseminate information about changing operational conditions to trading partners.
	Benefits of Channel Management	(ii)	Technical, product and pricing information required, repeated telephone calls and innumerable number of labour hours. Instead electronic bulletin boards are used allowing instant access. Electronically linking production with international distributor and reseller networks eliminates thousands of labour hours per week in the process.
(v)	Goals of Payment Management	(i)	To link the company and the suppliers and distributors so that payments can be sent and received electronically.
	Benefits of Payment Management	(i)	Increases the speed at which companies can compute invoices.
		(ii)	Clerical errors get replaced.
		(iii)	Lowers transaction fees and costs.
		(iv)	Increasing the number of invoices processed (productivity).

It is also a key to achieving maximum return on investment (ROI). According to AMR research, the planning and scheduling modules that depend on optimisation technology have generated 30% to 300%. ROI within companies that have already used them and the returns would continue to rise. Optimisation is a means to unlock the potential from all of the data residing in ERP in order to support better management decisions. It is the most exciting and promising segment within the supply chain management (SCM). This technology has moved into the spotlight because software and hardware breakthroughs have converged during a period when businesses are challenged to be more efficient, more effective and globally competitive.

At the heart of today's supply chain optimisation technology are complex algorithms that can examine millions of variables and solve increasingly complex problems in ever shortening timeframes, enabling solutions in a matter of hours rather than days. The knowledge and expertise to design such algorithms are in very short supply, however, residing within a small fraction of the world's software developers. It is not possible for all supply chain firms to be able to employ this valuable, scarce resource. To overcome this limitation, two factors are helping:

- Object technology (Helps to create the software component industry).
- Software components (These mainly handle routine functions such as database access).

The availability of extremely sophisticated optimisation components within the SCM industry now puts the most advanced technology within reach of both SCM application providers and companies designing inhouse SCM solutions. Many ERP suppliers have moved into SCM market. The biggest challenge for top ERP firms like SAP and J.D. Edwards was how to incorporate best-of-breed optimisation into their planned SCM products, since designing such technology usually requires more than five years of development work and extensive field experience. Many of these large companies have been able to achieve the time-to-market needed today by licensing intelligent optimisation components from front-ranked component vendors which cuts their development costs and gives their customers highly functional products in much less time.

Advanced Planning and Scheduling

The optimisation component suppliers being sought out by top ERP and SCM application vendors are those that deliver the most sophisticated algorithms and the clearest Advanced Planning and Scheduling (APS) within an intelligently constructed optimisation framework.

Industry experts believe Advanced Planning and Scheduling (APS) is the next evolution of manufacturing systems. Over the last three years, most APS products have been embedded with sophisticated optimisation logic. APS is a key part of an SCM application. APS is almost synonymous with supply chain optimisation. APS is considered to be one of the hottest segments in the entire realm of enterprise applications, primarily because of what its optimisation technology makes possible. Using its optimisation 'engines', an APS system synchronises the customer order workflow with the materials, manufacturing, and distribution activity, required to deliver the order.

Fuelled by its optimisation component, APS has quickly leaped to the forefront not just within SCM, but within the entire huge ERP market. For more than 30 years, ERP solutions had at their heart a Materials Resource Planning (MRP) system. Experts expect optimised APS to have a profound impact on the ERP market itself. AMR research recently predicted that APS would drive a \$1.4 billion (1998 estimate) market for applications to extend the \$ 15 billion ERP market.

Within the SCM arena, optimisation handles both operational and strategic decision-making. Users and vendors of supply chain optimisation believe that it is most useful in situations where a company or a product has:

- A complex supply base
- A complex manufacturing process
- A complex distribution system
- Volatile demand.

Supply chain optimisation is the best solution whenever there is uncertainty in the behaviour of supply chain operations or in market demand. Many users report that optimisation provides value regardless of the nature of the supply chain.

Supply chain optimisation can help those in various functional areas within a company take a broader, yet more effective approach to the whole supply chain, from raw materials to end-users. The decisions most impacted by supply chain optimisation are:

- Sourcing
- Make versus buy
- Production allocation
- Supply chain design
- Sales and operation

Operations is a prime area within which to deploy an optimisation solution, since the technology helps to identify and resolve problems like bottlenecks or unreliable suppliers. The software can suggest improvements and thereby support human planners faced with making difficult decisions. Any manufacturing company in which improving the supply chain could add significant value could conceivably benefit from optimisation.

12.8 CONSTRAINTS PROGRAMMING

A new technique to save a variety of planning problems called constraints programming (CP) was developed for problems in which the availability of resources is constrained and real-time scheduling decisions are required. Today, many APS systems are constraint-based. They consider the real-time availability of each resource before determining manufacturing and shipment dates. Optimisation is now performed by specialised software systems that are specific to various types of problems.

To understand the fundamentals of optimisation, a decision-maker must first understand the attributes of the business questions that need answers like:

- How many manufacturing plants are required? Where they should be located? Which products they should produce?
- How many distribution centres are required? Where they should be located? Which products should they stock and at what inventory levels? Which customer should the distribution centre cater?
- Which suppliers should we select?

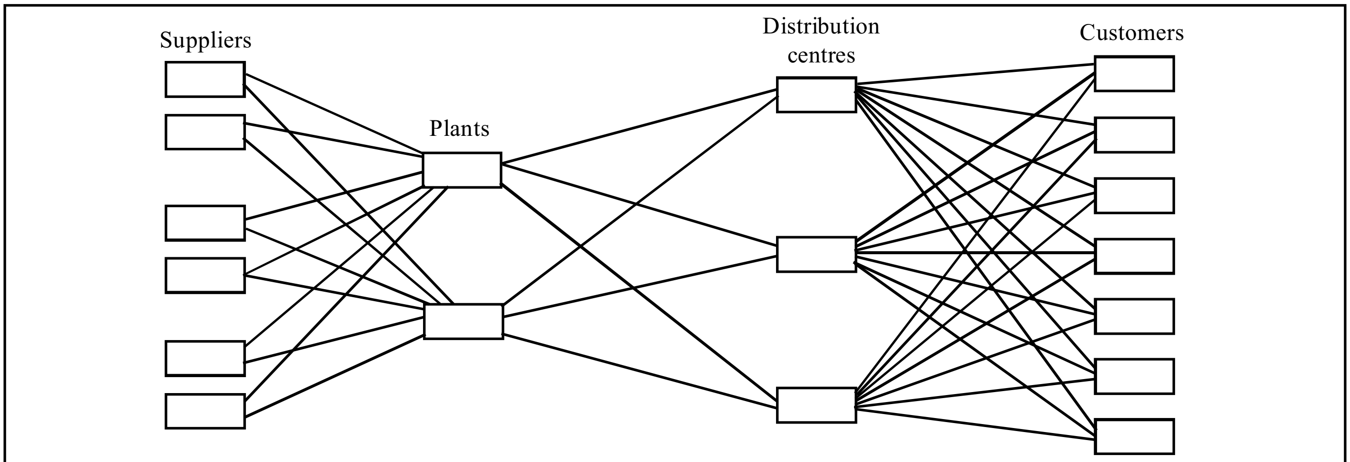

Fig. 12.7 – Supply Chain Design

When applying mathematical programming or constraint programming techniques to solve optimisation problems today, users must know how to model. They have to be able to write software to generate matrices and call a solver, and either use mathematical programming languages or write software in C++, using a solver or using a specialised computer programming language that support constraints programming.

Supply chain optimisation is the area within SCM that can most impact efficiency and deliver maximum ROI.

12.9 THE INTEGRATED VALUE CHAIN

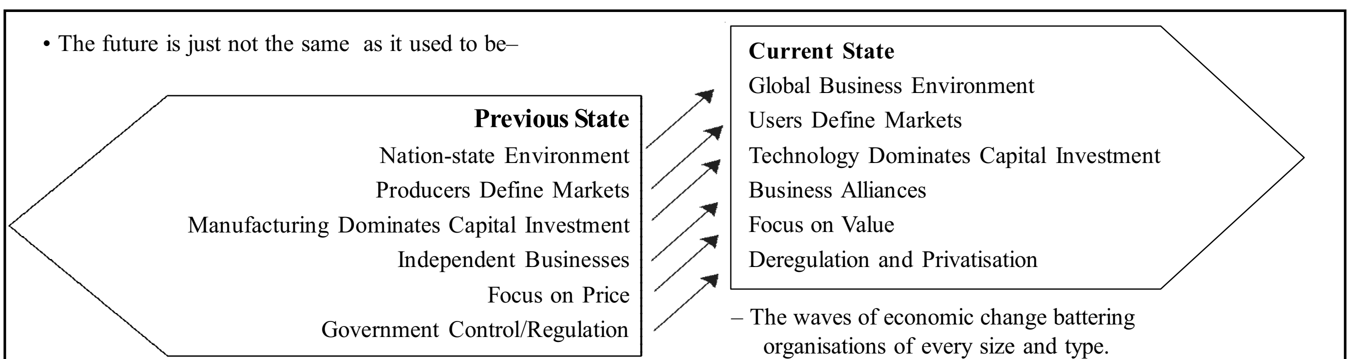

Fig. 12.8 – The Waves of Future (Economic Change)

This is now the competition's new battleground. Companies will fight the war for tomorrow's customers on a different battleground than in the past. In the earlier decades, business prized independence. The next decade will be on business alliances and competing, end-to-end value chains. All these are because of economic realities which are demanding that companies become customer-driven organisations that can identify and address customer needs from a variety of perspectives at every point along the value chain. Entire value chains in the future consist of powerful business alliance partners who will compete as single entities – eco–systems–for customers. These ecosystems will consist of collaborating firms that have entered into allowances and shared technology investments in order to win in the marketplace.

The Ground is Shifting

A series of simultaneous "power shifts" are permanently altering the rules of business. The ability to adapt to these shifts and transform them from problems to opportunities will separate winners from also run in tomorrow's economy. These business power shifts include:

To a large degree, dramatic changes in the financial, telecommunications, and transportation industries have propelled these power shifts. It would be difficult to identify a business unaffected by such shifts and especially by the changes that have put the consumer or business user more in charge of economic decisions.

Leading-edge companies looking for ways to offset these forces are now beginning to adopt a concept potentially powerful enough to blunt the effects of economic power shifts – the Integrated Value Chain. This concept will have a major impact, allowing companies, and ultimately customers, to benefit from reduced inventories, cost savings, improved customer service, and tighter links with business partners and suppliers. These value chains, and the organisations that comprise them, will constitute powerful competitive entities with compelling characteristics. They will be:

- Value leaders
- Technology/web-linked and quality
- Trendsetters in service and delivery
- Global in scope and/or reach
- Fiercely competitive on cost, service and speed
- Thought leaders

12.10 SUPPLY CHAIN – PRECURSOR TO INTEGRATED VALUE CHAIN

According to the Supply Chain Council, "Because manufacturing quality – a long-time competitive differentiator – is approaching parity across the board, meeting customer demands for product delivery has emerged as the next critical opportunity for competitive advantage. Companies that learn how to improve their supply chains will become the new success stories in the global marketplace."

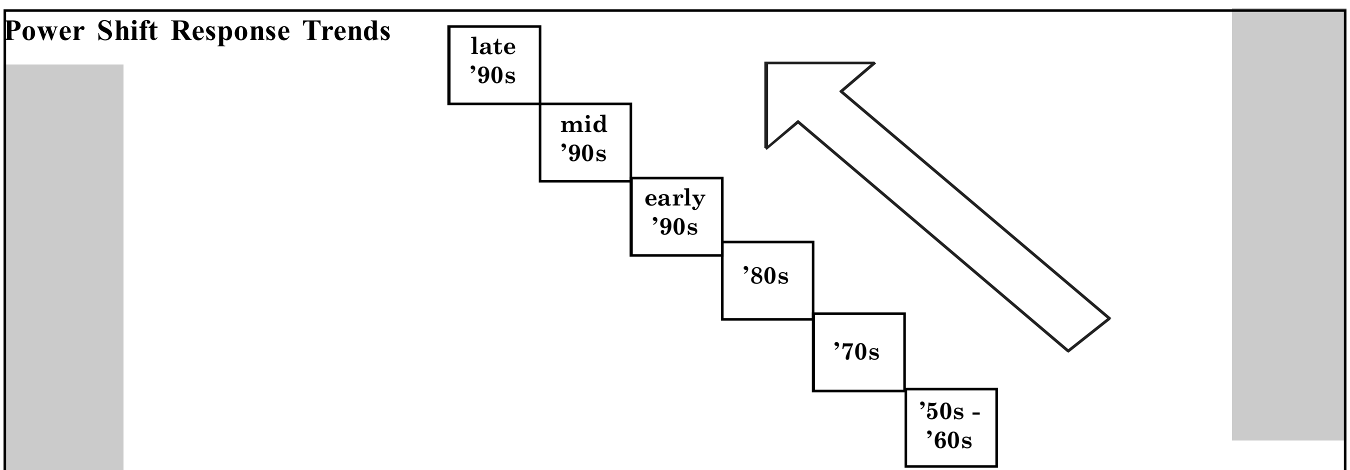

Fig. 12.9 – Power Shift Response Trends

While supply chain improvement is an important step towards the optimised value chain, back-to-front and front-to-back value chain integration requires not just improvement, but a new theory about how to create value for customers. The Integrated Value Chain concept acknowledges that the demand stream and the supply stream of a business cannot remain disconnected and still compete successfully with the likes of Procter & Gamble, Warner-Lambert, Microsoft, Home Depot, and Xerox. These companies and others are beginning to achieve competitive advantage by moving beyond supply chain to proactive value chain design and execution.

As on today, Oracle SCM (Value Chain Management) cloud delivers the usability, insight and capability needed to create client's own intelligent supply chain, best SC cloud solution to support clients unique need to accelerate growth. Also, it helps drive innovation and turn traditional supply chains into integrated value chains. Best-in-class companies are nearly 30% are likely to have an analytical disciplined process in place to execute planning and forecasting. Leveraging this kind of model is constantly evaluated and read to changes in the market is the difference between chaos and a smoothly operating supply chain.

A Difference in Scope and Intent

You'll come across many different definitions for supply chain, Integrated Value Chain, and new Business Ecosystems. Here's how they are defined:

Supply Chain – All activities involved in sourcing, producing, and delivering a final product, from the supplier's supplier to the customer's customer.

Value Chain – All internal and external processes that add value to a product or service (Michael Porter, Competitive Advantage, etc.).

Integrated Value Chain – A process of integrating and value enhancing all internal and external processes to increase the perceived value of a product or service.

New Business Ecosystem – An interacting community of business partners coming together under a single, leveraged brand to reach customers via the Internet.

But what is the Integrated Value Chain, and how does it differ from the supply chain? The two concepts differ in their scope and intent. The business world has traditionally thought of supply chain in terms of purchasing, transportation, warehousing, and logistics. Now, this concept has given way to the Integrated Value Chain – a larger, more encompassing concept. Value chain integration can be defined as:

"The process by which multiple enterprises within a shared market channel collaboratively plan, implement, and electronically as well as physically, manage the flow of goods, services, and information, along the entire value chain, from point of origin to point of consumption, in a manner that increases customer-perceived value and optimises the efficiency of the chain and thus creating competitive advantage for all stakeholders in the value chain."

Another way of defining value chain integration is as a process of collaboration that optimises all internal and external activities involved in delivering greater perceived value to the ultimate customer. A "supply chain" transforms into an Integrated Value Chain when it:

- Becomes highly customer-centric, focusing on demand generation and brand equity enhancement, as well as demand fulfillment and logistics.
- Includes the demand stream and the supply stream, encompassing the entire spectrum of a firm's business model/ process, from customer's customers to supplier's supplier.
- Is proactively designed by chain members to compete as an "extended enterprise," creating and enhancing customer-perceived value by means of cross-enterprise collaboration.
- Seeks to optimise the value added by information and utility-enhancing services.
- Enables creation of specialised value propositions for specific customers or customer market segments.

In this sense, the Integrated Value Chain concept applies equally to manufacturing and service business.

To emphasize the importance of this effort, Xerox has identified the Integrated Supply Chain as one of its four core business processes. In doing so, Xerox has reduced inventories company-wide by $1 billion, delivered logistics cost savings of $300 million, improved Return on Assets by three points, and improved customer service levels by 50%.

If these are the outward signs of value chain leadership, how do you recognise who the world-class firms will be at an earlier stage? What are the attributes that distinguish the companies, alliances, or value chains that will be the winners as we enter the next century? How do you become a value chain leader?

The top supply chain management companies in the world are:

- Unilever
- Procter & Gamble
- McDonald's
- Coco Cola
- Dell
- Cisco Systems
- Intel

The supply chain involvement is finding IBM, Watson, Lenovo, Master Lock and Eileen Fisher.

Create New Business Models and Specialised Value Chains to Increase Customer-perceived Value

Value chain leaders create new business models to offer a new way to deliver value to customers. These leaders focus first on the market and the customer, discovering how the customer wants to do business and creating new or improved “value propositions” accordingly. Businesses can no longer ignore the wants and needs of the marketplace and hope to survive. Someone will emerge to fill the customer’s unmet needs, often by marketing a product or service the customer didn’t even know what was possible. Customers – whether consumer-or business-to-business – will always act out of self-interest. They will behave in accordance with the following Perceived Value Model when making decisions to purchase or re-purchase:

$$\text{Customer perceived value} = \frac{\text{Quality} + \text{Utility}}{\text{Price}}$$

The Perceived Value formula explains why a well-satisfied customer, who was pleased with both the quality and price paid for a product or service, may defect at the time of the next purchasing decision, based on a perception of diminished usefulness.

Traditional Language	Value Chain Language
Mass Markets	Marketing/Customer Intimacy
Supplier Management	Continuous Replenishment Partnerships
Purchasing	Global Sourcing
Manufacturing	Lean Manufacturing
EDI	E-commerce/Secure Extranet
Transportation	Integrated Logistics
Customer Service	Customer Touchpoints and Self-service
Inventory Management	Vendor-managed Inventory
Build to Stock	Build to Order
Sales Management	Sales Force and Field Automation
Supply Chain Management	Value Chain Integration
Production Forecasting	Advanced Demand Planning and Forecasting
Vendor Competition	Supplier Rationalisation and Partnering
Warehouse Inventory Control	Demand-based Inventory Pull Strategies
Customer Satisfaction	Customer Loyalty
Annual Catalogs	Continuously Updated Electronic Catalogs
Cash Flow Management	Order-to-Cash Cycle Management
Customer Research	Data Warehousing and Data Mining

Table 12.4 – Traditional vs. Value Chain Language

12.11 VALUE CHAIN MANAGEMENT (VCM)

Value chain management, in the near future, is expected to go beyond supply chain management (SCM) to bring synergy between business partners by way of providing business and knowledge information in the most effective manner to help achieve business targets. It can be considered as a solution for easing out the interaction between various partners of an enterprise suppliers, dealers, bankers, etc. Three kinds of partners generally are involved against whom a company would like to build synergy.

(i) SCM Partners, Suppliers, Suppliers to Suppliers, Dealers, Customers, etc.

(ii) Transporter (transports, raw materials and finished goods)

(iii) Service providers and banks.

VCM covers all the above three types of partners and binds them. The focus is to connect all the points in such a way that there is a proper synergy between the partners. Further, the focus is on banker’s transactions to achieve the main objective of VCM of knowledge information sharing.

Business involves more and more transactions, and is not just mere debit and credit, material transfer, etc. If a supplier has to supply a component, the component manufacture details are to be known, i.e., to know proper specifications. Business partners should be made aware of the changes taking place for cost-effectiveness, value addition, etc. Such changes are usually communicated to the suppliers conventionally by drawings (on paper) or written processes (operation process sheets) or through CAD data on a cartridge. These are all nothing but knowledge transactions more than business transactions. Here, a third agency may be involved before the information really becomes useful to the supplier. The entire drawing data may be made available on the supplier's browser, in the same way that he sees his business data. Suppliers can view VRML files through available plugs-in without investment in high-end hardware or networking or software (VRML is Virtual Reality Modelling Language – a format for viewing 3D images). CAD (Core Company's Database) data is converted into VRML for real-time viewing. The business partners will come to know from the company about CAD data on:

- The products
- The processes
- How costing done?
- Quality norms
- New developments, etc.

The company is now in a position to work hand in hand with business partners after it becomes a real transparent organisation. The company will have a thorough interaction with their value chain partners through the Internet Platform.

VCM Architecture

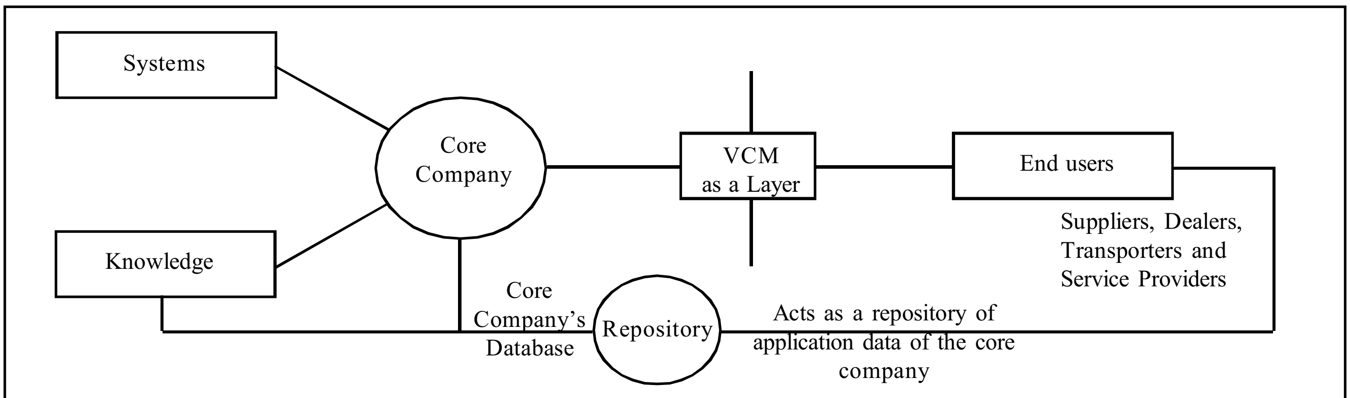

Fig. 12.10 – VCM Architecture

VCM is a layer between core company with its systems and knowledge on one side and its end-users on the other side comprising of suppliers, dealers, transporters, service providers, etc. who act as a complete repository of application and data of the core company and their partners together. This helps in bringing the partners together with the database of the company and application, and data from the suppliers to enable the companies repository to be updated for providing information to the end-users.

The architecture of the front-end solution should be such that it should be in a position to give the required information almost instantaneously with little access time. The data follows in a sequential way taken from the table structure and design. A drill down application such as an MIS (Management Information System) is system should have a table structure designed for that. A batch processing application needs a different table design. Independent of the back-end database, we need to restructure the database to meet the needs of the Internet user.

A separate database is needed for the reason that back-end databases can be a mix of ERP (Enterprise Resource Planning) and legacy systems and every back-end need not follow certain standards. We cannot rely on the back-end solutions to be compatible. Finally, this needs building up the value chain repository with our architecture, with certain amount of customisation, irrespective of the platforms being used. The other advantage is the security part. In the VCM arrangement, there is no possibility of any Internet user to come to the corporate ERP database. If the VCM database is corrupted, intentionally or otherwise, it will always be restored from the enterprise database.

The Interface at the end-user, will be with Internet browser, through interaction with their respective domains.

- Supplier domain
- Customer domain
- Dealer domain, etc.

The tools like e-mail, demos system documentation, help, etc. are built into VCM to enable users to get trained automatically. Two servers interconnected in fail safe cluster mode are needed, because if one fails, the other automatically takes care in just about 15 seconds. Adequate bandwidth is required when the users will start downloading VRML files. Connectivity to the Internet is either through radio modem or leased lines.

Availability of internet with number of ISPs, best option would be VCM. Comparing VCM with e-commerce, VCM is more of a core competence tool, whereas the latter is more of a marketing oriented solution. VCM talks about the business processes should be, covers quality standards, engineering information, etc. E-commerce is web-based business, whereas VCM is web-based IT.

VCM's Advantages

Savings by way of reduction in delays in transportation, uploading, despatch and preparation/transcription of documents, etc.

Acceptance of the challan, etc. on the Internet, the moment the track arrives at the material gate and the driver produces the challan, entire data gets transferred from VCM system to the material system. It is possible to produce a much better unloading plan from this.

It will also improve cash flows to the company, reduce communication costs arising out of strengthened relationship of the business partners.

VCM is a win-win strategy for all those who adopt this.

12.12 VALUE CHAIN LANGUAGE

Each new management technique has its own language. Value chain is no different, except that the language is more diverse and broader in scope. How frequently do you hear the language of value drain in your business?

Form collaborative alliances to strengthen and extend the value chain.

Increasingly, value chain leaders are turning to alliance-building strategies to strengthen and extend their value chains. Value chain alliances can take many forms, but fall under two broad categories:

Exclusive Alliances	Non-exclusive Alliances
Mergers	Marketing Alliances
Acquisitions	Joint Product Development
Joint Ventures	Cross-licensing Agreements
Supplier Relationships	Customer Relationships
	Technology Alliances and Linkages

Alliances play a vital role in many value chain strategies of all the trends that are sweeping across the business landscape. Strategic alliances may exert the greatest influence on business as companies race toward the next century. It views the business landscape not simply as a battleground pitting companies against each other, but as "a dynamic playing field where firms with complementary capabilities and common interests can coalesce to win competitive advantages".

Characteristics of SCM

(i) Sourcing of raw material or finished goods from any where in the world.
(ii) A centralised, global business and management strategy with harmless local execution.
(iii) Online, real-time distributed information processing to the desktop, providing total supply chain information visibility.
(iv) The ability to manage information not only within a company but across industries and enterprises.
(v) The seamless integration of all supply chain processes and measurements, including third party suppliers, information systems, cost accruing standards and measurement systems.
(vi) The development and implementation of accounting models such as activity-based costing that link cost to performance are used as tools for cost reduction.
(vii) A recognition of the supply chain organisation into high-performance teams going from the shop floor to senior management.

12.13 ELECTRONIC LOGISTICS AND ITS IMPLEMENTATION IN BUSINESS HOUSES

For those netizens who think that the internet is used only for sending e-mails, or merely for surfing through colourful sites, here's an interesting news! If you haven't heard of enterprise resource planning (ERP), or internet-enabled logistics (E-logistics), you haven't heard of anything yet!

Today, more and more companies are making use of ERP and extended ERP (e-ERP) in order to improve their business transactions.

What is E-logistics and How is it Implemented in Business Houses?

In the US parts of Europe and in Singapore, a robust IT infrastructure has been built up that includes the Internet, and local intranets. But this infrastructure also needs to be used for the right kind of applications. Businesses are beginning to use the internet for their cost and time advantage. Currently, huge amount of inventory is being kept in factories that is both unwieldy to maintain and creates delivery time problems. So, the question is: can information substitute for the inventory? By having information on the number of items to be sold the next day, a business house can keep the right amount of inventory. This is achieved through information sharing, customer e-mail, scanning, customer surveys, etc. Such online information is collected and further used to a business advantage.

In addition to information collection, the physical movement of components and goods along the final manufacturing value chain is also important. Movement of material from one business to another is a logistics problem. So, in addition to the IT initiative, logistics initiative is also required.

There is tremendous potential for India. Several companies in India are component manufacturers to foreign equipment companies. Also, software companies develop component software on a contract basis. Knowingly or unknowingly, several of these companies are part of the global manufacturing enterprises. Several orders of magnitude of cost savings is possible with appropriate logistics, inventory control policies and use of IT tools. For example, India has several sea ports and airports. But one needs to study the facilities and operations from the logistics point of view and improve them.

In India, we have the IT initiative, the infrastructure initiative and the telecom liberalisation from the government. Do these contribute in any way to India's competitiveness in the international arena?

All these initiatives are long overdue steps in this area. For initiatives to be converted to action plans, it requires R&D and the hardwork of the intellectual community. If you consider the infrastructure initiative, an international airport is 50% concrete and 50% IT. It is simply an IT-enabled enterprise moving passengers, their baggage and more importantly, air freight. An airport has warehouses which require management. All these require sophisticated optimisation models and monitoring software. One has to build Internet-enabled information systems, incorporating state-of-the-art optimisation and project scheduling algorithms. The key to all this is Internet-enabled logistics.

It is basically transaction processing software, enabling millions of transactions to be systematically handled. Auto networks, food enterprises, garment networks and several other businesses use the 'extranet' for information and funds transfer.

Logistics is the function of moving the right material to the right place at the right time in a global enterprise. Transportation takes time, inventory, money. If you take the cost of the car manufactured, 70% is incurred in transportation, 30% material cost. 10% time is used for assembling the car and 90% in getting it to the customer. If you get cost-time advantages of a product, it is on the roads 90% of the time. This is pipeline inventory. To make business efficient, one needs to cut down inventory and logistical costs.

It is possible through the use of the internet and supply chain planning. Let's say a TV manufacturer needs components like the tube. If a supplier provides at the right time just the right number of components, then, at the end of the day, only the finished product is present with no idle components left. This is both cost-effective and competitive. To do this, the manufacturers have to communicate with component suppliers online. If suppliers have the manufacturers' production schedules, they can tune their own production based on this information.

There are what are called exchanges. Manufacturers and suppliers are connected through 'extranet'. Information is therefore firewalled. Several encryption keys exist to ensure security. Take a site like Amazon.com. You can select a book, place an order, give your credit card number, and ask for home delivery. Transactions are electronically taking place, except the physical delivery. All this is taking place in a short time. If Indian companies streamline their delivery process using the existing IT infrastructure, they can make improvements. If businesses start using and paying money to internet, infrastructure improves too. Banks also must cooperate.

organized marketplaces (often online) where buyers and sellers meet to trade goods and services efficiently.

Basically, a company can have several kinds of data that is scattered (supplier data, personnel data, finance data, etc.). Computers holding data don't talk to each other. In ERP, this information is integrated, stored on a single database. A single query therefore can give a host of information. The beauty of ERP is information integration. Today, there is an integration of various companies. You may manufacture the best product in the world, but if you cannot supply it to the customer on time, the company winds up.

No one company can do everything. Each has its own 'core competency'. So, different companies have to come together (from raw material to assembly to transportation to delivery), to serve the customer. Amazon.com does not publish books. It only delivers them. Publishing houses cannot engage in sales online. But combine the two, your capability increases. Joint companies on the value chain should join together and iron off their interfaces. Business improves vastly. Competition is between value chain, not between companies.

This is organisational trust. When companies trust each other and share information through a communication network, they improve performance by several orders of magnitude. It has created new research areas like risk analysis in information sharing, etc.

Transactions between companies can be facilitated through inter-organisational information systems. If there are two organisations, a bank and a manufacturer, we need to design methods to transfer valuable and helpful information. A component manufacturer must understand what time delivery must be made, where it is to be used, etc. A lot of algorithms go into designing this system.

Transactions are further improved through data warehousing. If you take a supply chain of a product, you need to know your customers. 80% orders are said to come from 20% customers for 20% products. So, companies must find out which products are required and where.

I believe that a revived area of traditional research is forecasting. This is predicting the demands of customers. Statisticians look at forecasting as a new area that can compress lead time.

There are several reasons why enterprises work. A company does not enter the network alliance unless it finds some value addition. Conversely, no company is allowed to enter a network unless it can add value to the network. This is because competition nowadays is not between individual companies but between company networks. Also, the network governance structure is such that one of the companies takes leadership role and others cooperate.

12.14 SUPPLY CHAIN PORTAL (SCP)

Supply chain e-markets help the enterprise enact a recombinant business model.

- During the first half of 2001, a spate of supply chain e-markets will emerge run by the equivalent of shared service providers.
- Instead delivering a highly integrated suite of functions focussed externally say ERP, these portals will deploy narrow functions across multiple businesses, regardless of location.
- Portals will use browser graphical user interfaces (though certain types of application will use customised front-ends).
- To facilitate links with other application types, they will use message-broker middleware technology and have a service-oriented architecture.
- They will integrate with external systems (from the inside out, instead of outside in).
- They will accommodate, rather than exclude data, processes and applications.
- Those seeking to sell supply chain e-markets will include:
  - Value-added networks
  - Whole salary
  - Channel masters
  - IT vendors
  - 3 PLS
  - Systems integrators, etc.

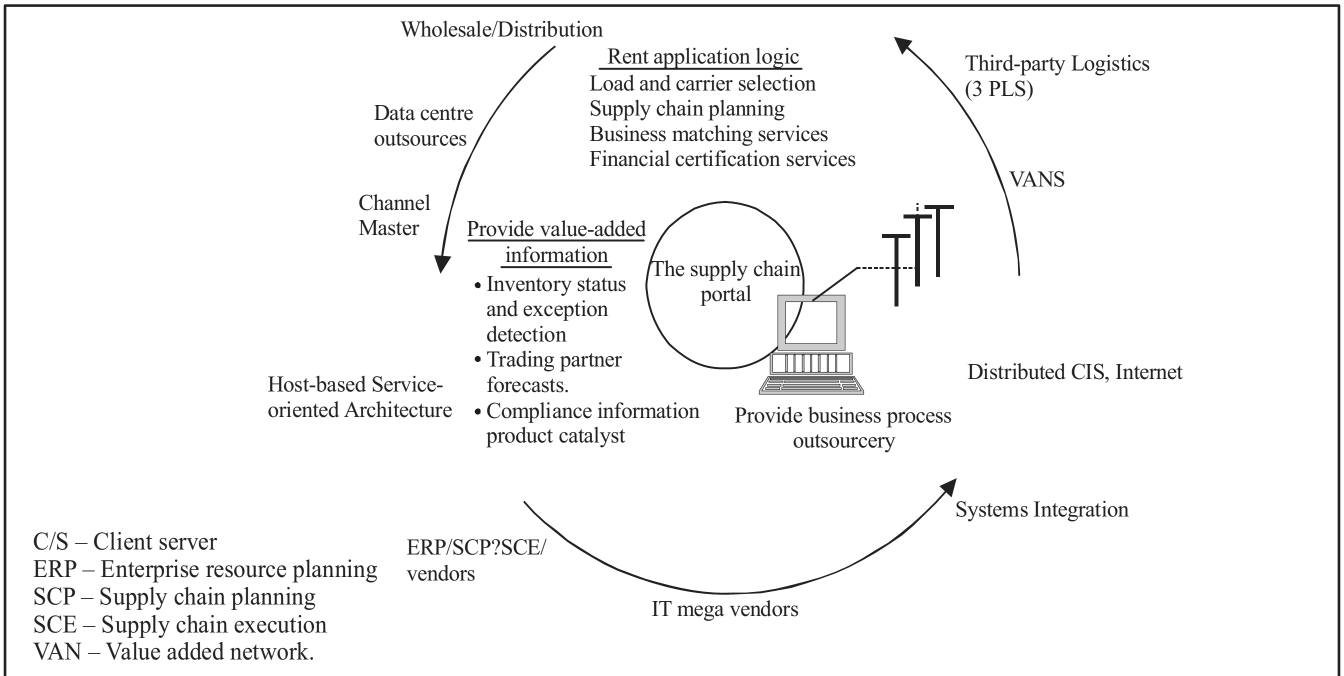

Fig. 12.11 – The Supply Chain Portal

*[**Source**: Gartner Group]*

- The more successful providers will evolve to become e-markets for specific supply chain by providing value-added information and business process outsourcing service on top of simple access to an enterprise application or shared community data.
- Profitability – should use emerging tools for optimising financial returns to make SCM programs key drivers of financial performance.
- Enterprises should focus on the following basic principles:
  - Connectivity – should create an integration architecture for SCM (Supply Chain Management – internally and externally.
  - Agility – should recognise that profitability depends above all on agility, requiring a new breed of systems that plan, monitor and read; requires supply chain structures.
- Higher level of the enterprise becomes involved in discussions on SCM strategies. Too often, these high level initiatives focus on buying the 'strategic application solution' with no appreciation for the complexity and immaturity of SCP and SCE. (Supply Chain Planning and Supply Chain Execution)
- ERP vendors also considered as enterprises "strategic application vendor" are engaged in catching up in SCP and SCE software functions.
- Enterprises should know that success with SCM depends on speed at which they work. It is similar to enterprises' business success in the year 2005 where it is anticipated that success will depend on its agility. The success of its SCM strategy will depend on how quickly it is implemented and on its ability to meet new business demands.

Various Companies	Offers (SCP and SCE Software)
1. Trade matrix.com from i2 techno	Multidimensional trading technologies and various partners. Partner communities (Initially focuses of SCP collaboration).
2. i2 partners with other software (They work for other companies by taking links)	Expanded functions like in vendors (Right procurement, pricing and logistics.
3. Bstream2.com	Integrates distributors, channel masters & work with other enterprises.
4. MySAP.com (from SAP)	(This initiative emphasises transactions and information exchange; Future plans include a focus on supply chain collaboration).

5.	Owens & Minor	A distributor of medical and surgical products, allows trading partners to analyse historical performance and connect with trading partners.
6.	Capstan for International Trade	Provides hosted software to software, handle import and export control and inventory visibility.
7.	General Motors Trade Exchange	Procurement and hosted SCP solutions.

- Waiting for the 'strategic enterprise vendor' to deliver generic SCP and SCE functions will ensure that an SCM strategy differentiates, the enterprise competitively little (if at all). If the enterprise seek to gain business advantage through SCM processes, they must clearly identify where they must standardise their business process and which they must differentiate those processes. They should assemblies and manage their application portfolio accordingly.
- A dangerous misconception shadowing the SCM market exists that since ERP vendors have decided now to offer SCP functionality in addition to transactional functionality. ERP vendors have succeeded in the battle for SCM.
- SCM is involved in optimising the delivery of goods, services and information from the supplier to the customer. Instead of considering this as a specific technology, SCM is a philosophy for improving the enterprise's competitiveness within the context of the markets served by that enterprise. Competitiveness is a continuously rising phenomenon.
- Applications that enable SCM philosophies (SCP and SCE applications) have just begun to scratch the surface of what can be achieved with powerful computers, advanced integration technologies, and innovative solving methods.

SCP Tools

The supply chain planning tools (SCP tools) are divided into three groups:

- Strategic planning tools
- Tactical planning tools
- Executive or implementation tools

Strategic planning tools covers the following:

- Supply chain management strategy.
- Customer service policies.
- Channel design
- Network design.

Traditionally, enterprise have used strategic planning tools periodically as stand-alone solutions. In the recent years, as e-commerce has involved fast, business relationships and channel strategies have become more dynamic and sensitive. Enterprises should integrate strategic planning tools tightly with tactical SCP, sometimes including information and processes from trading partners.

As Gartner reports put SCP as a subset of SCM and is the process of coordinating assets to optimise the delivery of goods, services and information from supplier to customer, balancing supply and demand. An SCP suite sits on top of a transactional system to provide planning, what-if scenario analysts capabilities and real-time demand commitments. Typical modules of tactical planning tool includes:

- Network planning
- Available to promise (ATP)
- Capable to promise (CTP)
- Capacity planning
- Demand planning
- Distribution planning
- Manufacturing planning and scheduling planner
- Deployment planning
- Transportation planning and scheduling
- Sales and operations planning.

– Available to Promise (ATP) is uncommitted inventory and planned production in master scheduling to support customer order promises.

– Assemble to Order (ATO) is a strategy allowing a product or service to be made to specific order, where a large number of products can be assembled in various forms from common components. This requires sophisticated planning process to anticipate changing demand for internal components or accessories while focussing on mass customisation of the final products to individual customers.

– Capable to Promise (CTP) refers to a system that allows an enterprise to commit order against available capacity as well as inventory. These systems are evolving to include multiple sites, as well as the entire distribution network.

Integrating structure planning tools with tactical SCP enables enterprises to set customer service policies, spares and repairs rules, sourcing guidelines, and other business rules as condition and strategies change and then automatically pass these guideposts down to the tactical planning tools.

Tactical planning tools receive much attention as on date. But enterprises must expand their focus if they like to improve the return on capital employed from these applications.

Few tactical SCP systems receive accurate real-time information about what is really happening on the plant floor, at the loading dock or transit. In the absence of this information, the enterprise optimises plans and schedules around the wrong constraints and suffers missed deliveries and uses assets inefficiently. To improve tactical operations in manufacturing and distribution operations, enterprises will have to know how these improvements will affect their physical network. The enterprise may also have to rethink how it serves its customers and from where. The enterprise may also have to change its physical network to support fundamentally new processes – like outsourced manufacturing and postponement (providing a unique configuration for a customer's good, right before shipment).

Strategic Planning Tools

For complex enterprises, successful supply chain management depends on the enterprises's ability to create an IT architecture that passes data and business policy for every event between strategic planning, tactical planning, and supply chain execution (SCE) systems.

Capable to Promise (CTP) Systems

CTP has already been defined earlier, and on the contrary to ATP, CTP systems enable an enterprise to commit orders against available material and capacity in real time – in view of this, the enterprises do not have to hold excessive quantity of focussed goods in inventory. A CTP system also allows the sales organisation understand the consequences of accepting an order such as the order accepted rate would affect the other earlier orders already committed and what additional expenses would make the order unprofitable, etc.

In order to help enterprises make fulfillment more reliable and to use the assets effectively, vendors are adding functions to CTP systems that address resource allocation, deliverable and profitability. Many of the enterprises have adopted CTP systems to advise them on what products are to be made and shifted. In addition, they have added deliverable to promise systems: (i) to choose rates and routers accurately and dynamically, (ii) to offer load discounts, (iii) to give expected delivery time in hours and minutes (instead of only the shipment date) and (iv) the expected lead time in days or weeks.

Order management personnel, with the help of front-end traded deliverable-to-promise capability, can significantly increase revenue while improving the customer's experience and providing sufficient information to them and to enable them to make intelligent choices. A company can indicate to its customer's accurate delivery times coordinated with the customer's receiving windows, while allowing that customer to choose to receive the product (say from a truck) at a discount vs. several less-than-full shipments.

Research organisations have felt that enterprises that fail to model their supply chains accurately and to give the sales organisation access to capable-to-promise (CTP) information will lose 15% of potential profits from transactions because they will promise their customers too little or too much. Enterprises who are used to available-to-order (ATO) and made-to-order (MTO) manufacturers should have CTP systems as a top priority since these would improve customer service as well as transactions profitability while lowering operating costs.

Enterprises should integrate CTP systems into the segments of the order taking process, including web commerce, high volume phone orders and sales configurators.

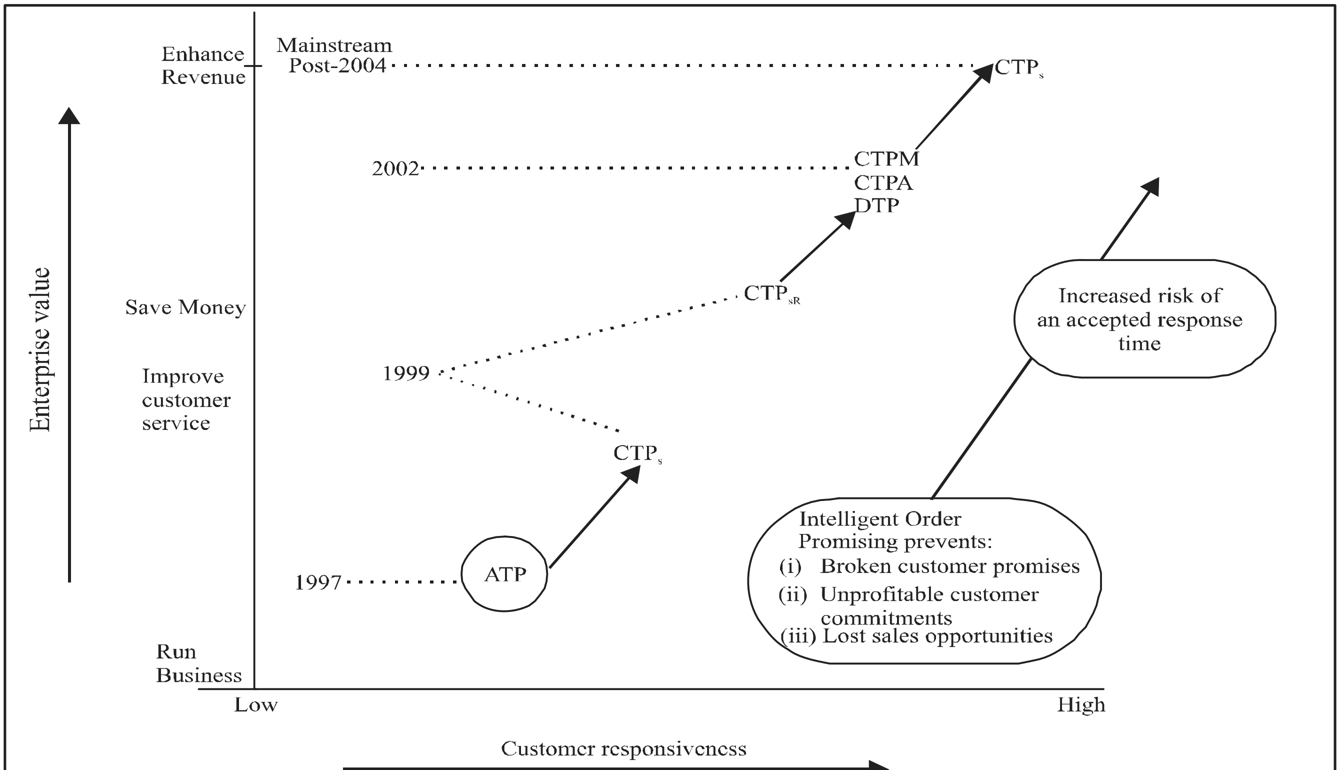

Fig. 12.12 – Realtime Order Promising (99% Accuracy Required)

Legend

CTP	––	Capable to promise
ATP	––	Available to promise
CTPs	––	Single-site CTP
CTP_{SR}	––	Single-site CTP with rescheduling
CTP_M	––	Multisite CTP
CTP_A	––	Allocated CTP
DTP	––	Deliverable to promise
CTP_F	––	CTP with financial optimisation.

Collaborative Planning Initiatives

Collaboration will be the name of the game for work process in the future. As the need to become physically located right next to the work area becomes less important due to the internet, the need for collaboration increases. Most of us have been prepared for this transition as it has just happened to us. Unlike our children who have grown up with chat and instant messaging, the workforce is still adjusting to the breakdown in order and structure that the Internet and the collaborative technology brings. The internet is creating new work process opportunities to build new networks of partners and suppliers in a very short period. For many businesses, selection of partners and how you work with them takes several months and lots of due diligence to make a decision. However, with the Internet and a flexible range of services, many companies can repackage their service and product offerings to add them to others and make new products or services.

Enterprises' reluctance to rethink in doing business with trading partners continues to constrain Internet-enabled collaboration. In this field, vendors like Syncra Systems, i2 Technologies, Logility and Manugistics have started to gain traction with their collaborative planning, forecasting and replenishment solutions, adoption of collaboration in most industries. However, this will remain slow as enterprises establish the rules of engagement with their trading partners.

Collaborative planning initiatives will gain momentum in the latter half of the year 2000, when enterprises tries to switch from monitoring suppliers' year 2000 compliance to looking for ways to improve margins and better satisfy customers through collaboration with supply chain partners. The initiatives in the earlier stages will start with only a few trading partners and will expand as enterprises establish trust and determine ROI (Return on Investment). The high tech industry is already geared up to deploy rapidly Internet derived technology to share information and to collaborate. In view of industries focus about high-speed, virtual manufacturing, enterprises must either collaborate or lose significant market presence.

Enterprise collaboration system provide tools to help us collaborate to communicate ideas – resource sharing and coordinate our cooperative efforts as member of the many formal and informal process, project teams and workgroups that make up many of today's organisation.

The goal of enterprise collaboration systems is to enables us to work together more easily and effectively by helping us to:

Communicate : Sharing information with each other.

Coordinate : Coordinating our individual work efforts and use of resources with each other.

Collaborate : Working together cooperatively on joint projects and assignments.

Collaboration is the key to what make a group of people a team, and what makes a team successful.

Four highly visible, mostly complementary functions provide or support collaborative supply chain technologies including:

- SCP vendors
- Managers of trading partners' extranets

  (Example:- General Electric Trading Information Exchange, Science Applications International in the automotive industry and its Scoptima – a solutions in high tech.)
- Industry standards groups

  (Example: UGC Net in retail and consumer goods, Rossetta Net in the IT supply chain, Bolero Net in the international documentation and documentary credit management market).
- Large IT infrastructure vendors

  (Example: Microsoft with its Biz Talk initiative)

Enterprises should launch pilots to learn first hand, the technical and business challenges – and rewards, with the goal of achieving e-commerce interaction to obtain competitive advantage. C-commerce refers to collaborative, electronically enabled business interactions among an enterprise's internal personnel, business partners and customers throughout a trading community who could be an industry, industry segment, supply chain or supply chain segment.

Supply Chain Execution (SCE)

It is a subject of SCM. This is also a framework of execution-oriented applications that enables the efficient procurement and supply of goods, services, and information across enterprise boundaries to meet customer-specific demand. In a broader sense, SCE includes the manufacturing execution system (MES), warehouse management system, and other execution systems within the enterprise, as well as throughout the supply chain. The logistics-oriented elements of SCE include the transportation management system, warehouse management system, international trade systems (ITSs), real-time decision systems (e.g., dynamic routing and dynamic sourcing systems), and supply chain inventory visibility systems.

Supply chain inventory visibility (SCIV) are the applications that allow enterprises to monitor and manage events across the supply chain to plan their activities more effectively and pre-empt problems. SCIV systems enable enterprises not only to track and trace inventory globally on a line-item level but also submit plans and receive alerts when events deviate from expectations. This visibility into orders and shipments on a real-time basis gives enterprises reliable advance knowledge of when goods will arrive.

The Internet forces enterprises to work directly with both suppliers and customers to respond rapidly and intelligently to change.

The pressure of e-commerce (particularly e-markets) will reveal serious weaknesses in the physical supply chains systems and processes (e.g., order processing, logistics and customer service) that interface with those of customers and suppliers, (e.g., shipping cannot be a stand-alone process but part of an integrated business process).

The strategic planning assumption is that through 2005, Web e-commerce applications will lead to missed delivery commitment dates, excess inventory and incomplete orders for enterprises that do not invert in back-end fulfillment applications and process integration (This is with 0-8 probabilities).

Enterprises striving to lead the market must use Internet technologies in conjunction with SCM applications:

- to bring new products to market faster
- to create faster fulfillment processes
- to increase their own ability to satisfy many different customer bases.

IT technology promises real-time visibility across multiple enterprises. Many more enterprises can employ strategies such as the global shop floor and the virtual enterprise.

Enterprises pursuing web commerce strategies must realise that success depends on fulfillment in a dynamic environment, which in turn requires a new breed of SCE systems. Emerging 'SCE markets' obscure that most SCE offerings today just repackage traditional warehouse management system (WMS) and transport-management system functions (TMS). Warehouse management system is a software application that manages the operations of a warehouse or distribution centre. Application functionality includes:

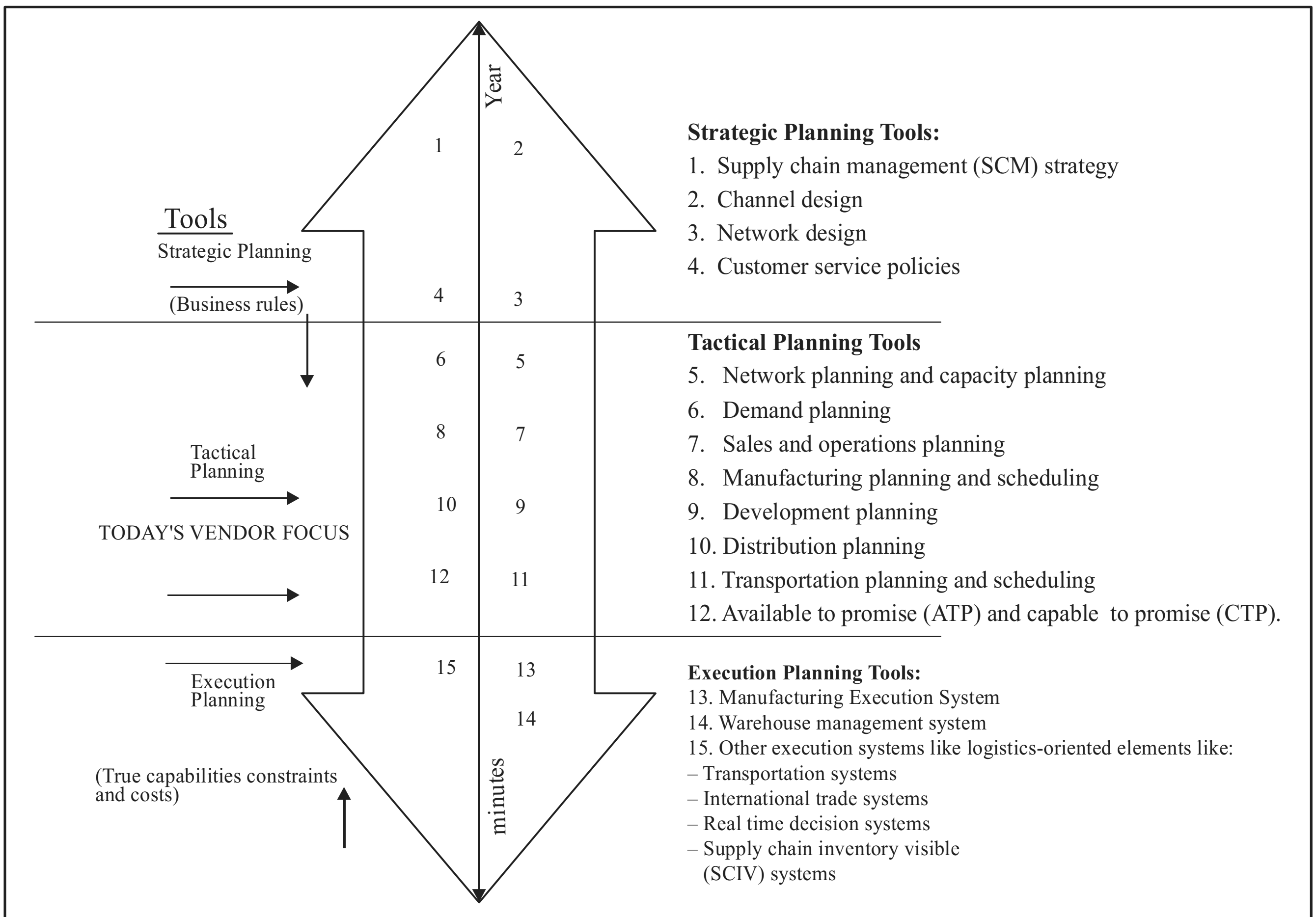

Fig. 12.13 – Supply Chain Management Disconnects

[*Source:* Gartner Group, RAS Service, R-10-0274, 2nd March, 2000].

- Receiving
- Shipping
- Order allocation
- Inventory management
- Cycle counting
- Automated material handling equipment interfaces
- Packing
- Wave planning
- Putaway
- Labour
- Order picking
- Replenishment
- Task interleaving

The use of radio frequency technology in conjunction with bar codes provide the foundation of a WMS, delivering accurate information in real time.

Transportation Management System (TMS) is used to plan:

- Freight movements
- Do freight rating and shopping across all modes
- Select the appropriate route and carrier
- Manage freight bills
- Payments.

Many enterprises implemented TMSs to achieve enterprise-wide load control centres although most applications failed to provide the necessary capabilities. To address the complex requirements of transport, vendors produced planning applications that use sophisticated algorithms to choose the best shipping scenario. However, transport planning and execution systems still did not satisfy requirements for enterprise-wide shipping as these resulted in relatively static point-to-point solutions. In response, the vendor community began acquiring TMS companies to offer a new type of integrated WMS and TMS known as SCE.

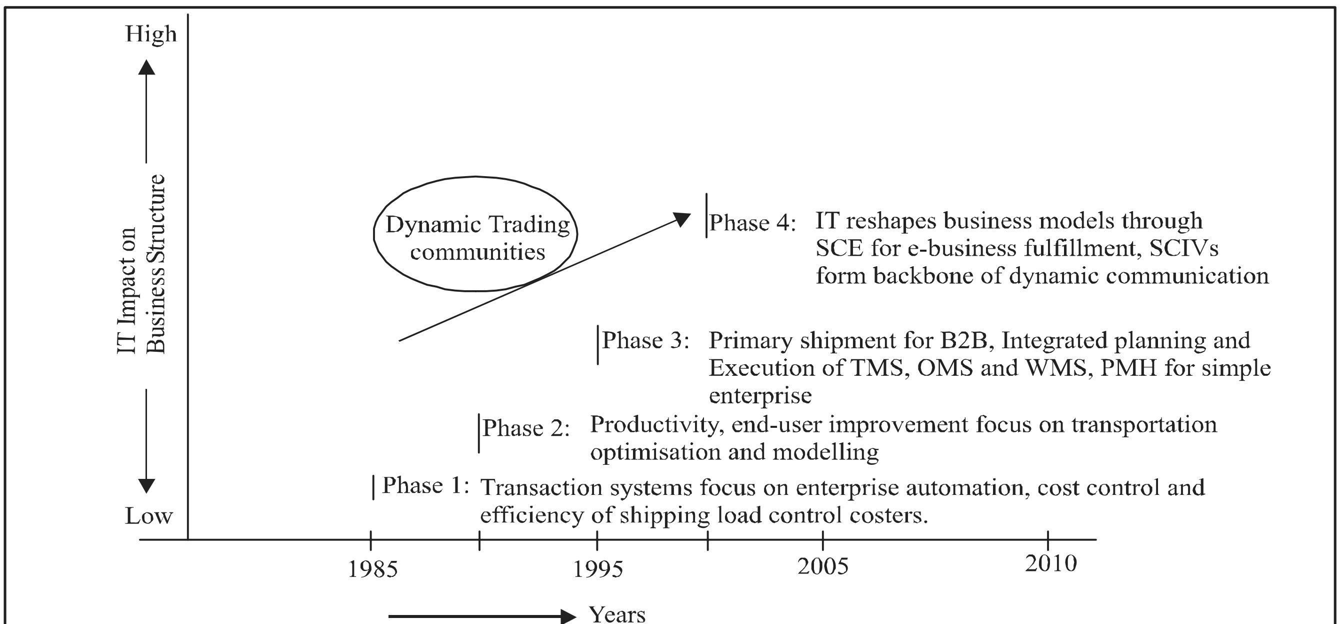

Fig. 12.14 – Supply Chain Management Disconnects

(***Source***: Gartner Group, R-10-0274, RA Services, 2nd March, 2000)

B2B	Business to business
OMS	Order management system
PMH	Product marshaling and handling
SCE	Supply chain execution
SCIV	Supply chain inventory visibility
TMS	Transportation management system
WMS	Warehouse management system

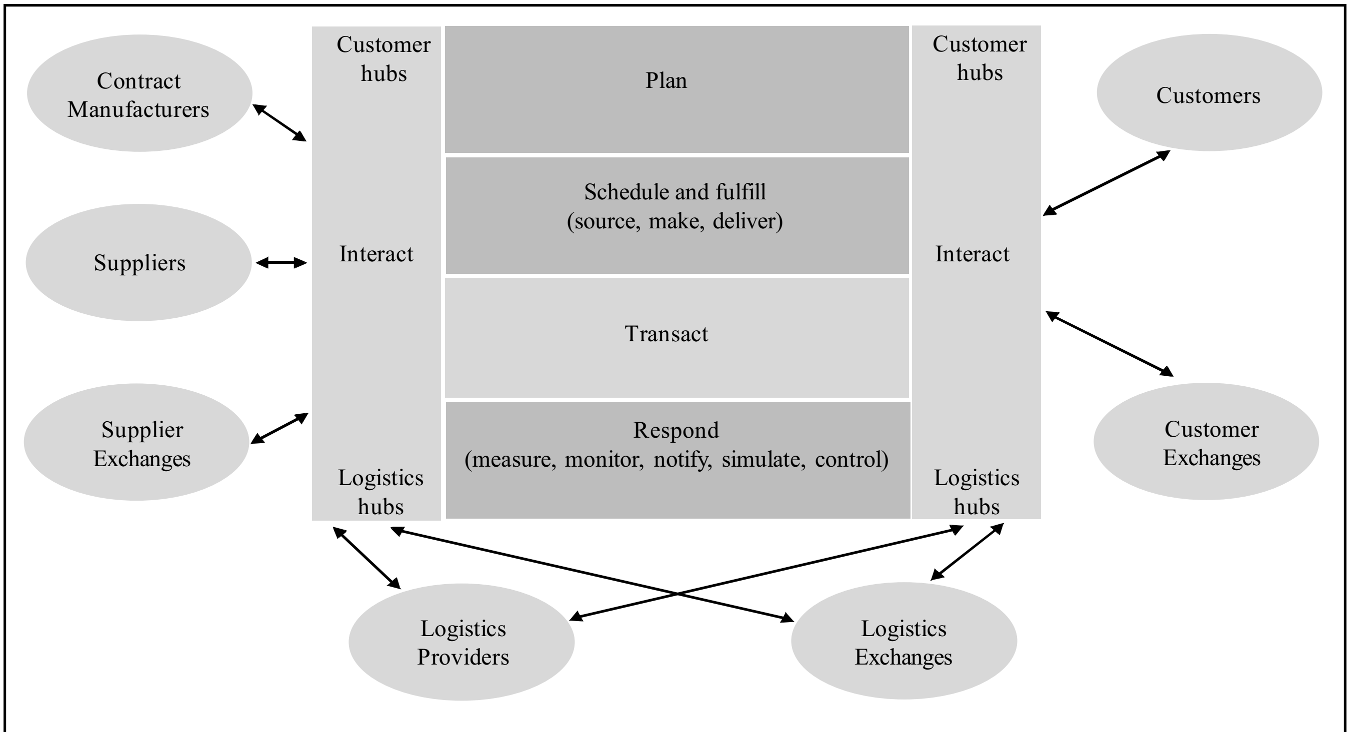

Fig. 12.15 – SCM Enterprise Business Process Model

As supply chain communities continue to expand, companies want to maintain control, but at the same time, provide flexibility and responsiveness to other trading partners. One way they are reaching these goals is by connecting to virtual marketplaces and trading exchanges.

Today's SCE systems fail to meet the need for systems that focus on customers and operate globally. The WMS/ TMS incarnation of SCE will likely be obsolete by 2005 as these systems create redundant processes, systems and data, thereby increasing inventories and showing reaction times.

The concept of integrated TMS, WMS and order entry management systems looks compelling on the surface. But they have shortcomings that keeps them for achieving the true SCE. This type of PMH does not optimise execution in real time or even via integrated systems, nor does it have systems to make all supply chain inventory visible to the enterprise in real time. This visibility forms the backbone for SCE while real-time decision support systems must be developed from transport planning systems.

SCE needs traditional combinations of algorithms and a new class of techniques, suitable for incremental planning with shortened planning horizons – for optimising executions. True SCE functions are evolving that will make the enterprises meet the customer's changing demands reliably, responsively and cost-effectively.

SCE system has been defined by Gartner Group as a framework of application that support the execution of supply chain processes to enable the procurement and supply of goods, services and information across enterprise boundaries to meet the demands of specific customers.

The component applications will come from several vendors but will follow a common architecture that lets transactional data, business logic and decision support information about individual customers and their orders pass between components in real-time. This setup enables a closed-loop business process. (These little human intervention) reduce cycle times.

Example: A database might take:

– a truck's location
– link it with planned shipment data
– link the truck to a production schedule
– load configuration
– route
– weather
– delivery constraints, etc.

In order to increase agility and meet changing requirements, the SCE system must provide an interoperable, component-based messaging architecture to allow rapid modification of new business partners or functions.

When globalisation of markets and proliferation of distribution networks takes place, ERP and SCM vendors will incorporate into their applications functions to manage international trade systems. Due to the complexity of each country's rules and processes, much content and domain expertise are to be dealt by the vendors specialising in international trade and in some large international third-party logistics firms. Inter-enterprise data sharing and documentation will develop – to support duty drawbacks as well as preferential tariff treatment programs. Vendors will be able to support:

- Automated customer cheques
- Subsequent transaction checking for compliance with export controls
- Harmonised code classification
- Consistency with World Trade Organisation valuations
- Duty, value-added taxes and other tax engineering opportunities, etc.

Net and total landed cost will become a standard key performance indicator (KPI) for international transactions. Big enterprises now manage these functions as a compliance cost. As more robust international trade-systems, functions and capabilities develop, more enterprises will use them as background functions in their order management and SCE applications. Enterprises with high-speed and high-variability processes should invest in SCE applications to increase customer service and profitability.

SCE Framework

The SCE framework is as shown below in Fig. 12.16 Traditional ERP functionally is shown at the top of the sketch which are warehouse management, transportation management, order management, purchasing, inventory management and financials.

Decision support for SCE requires different approaches and technology in contrast to SCP applications. Both SCE and SCP systems answer different questions.

The following will enable recombinant business models:

- Supply chain e-markets
- Uniform transport
- Semantic standardisation and reconciliation
- Consequential interoperability framework
- E-markets and Internet agents
- Common company and product identifiers
- XML name spaces
- Distributed object computing

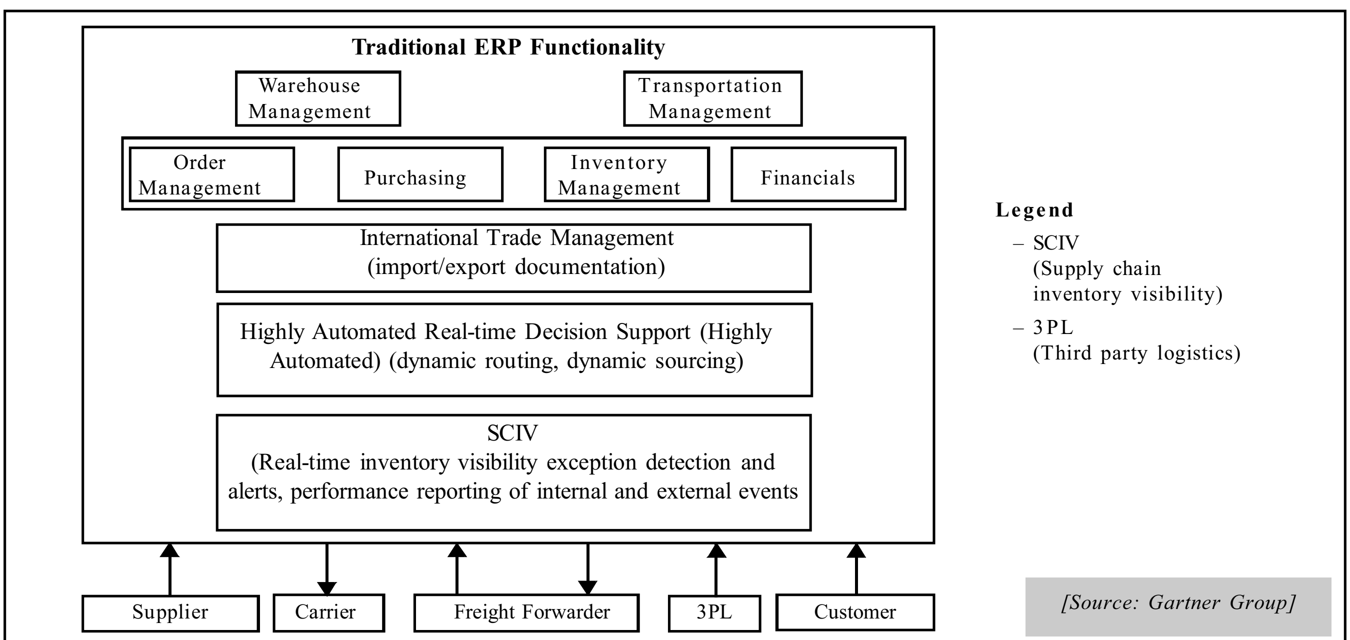

Fig. 12.16 – SCE Framework

SCP	SCE
• Planning based on expected conditions.	• Decision support based on actual conditions.
• Solution space shrinks as planning becomes more granular.	• Solution space grows as events deviates from plans.
• Vendor focus.	
• Point of execution customer commit point	Point of execution: Customer commit point.
Characteristics:	Embrace variability.
Goal: Minimize variability	"Real" event-based decision support.
Statistical-based planning	Fast, feasible algorithms.
Optimisation algorithms	Highly automated decisions.
Interactive scenario planning	Mandatory closed-loop process.
Optical closed-loop process	

Table 12.5 – SCP/SCE Comparison

12.15 INTERNET'S EFFECT ON SUPPLY CHAIN POWER

Three popular theories exists as under:

(i) *Theory One: The Internet will level the playing field:* The Internet, combining with the emerging global logistics network, will reduce the brick-and-mortar advantages of many traditional operations, thus leveling the playing field for all channel members putting the large at risk by the amassed might of the networked small.

(ii) *Theory Two: The Internet will cause middlemen to disappear:* The Internet will let large manufacturers deal directly with end customers. The losers will be distributors and retailers, the middle layers that add cost and elongate cycle times, and which no longer have exclusive access to end customers. The manufacturers will exert dominant forces in a truncated supply chain, perhaps teaming with third party logistic firms to provide final delivery to customers.

(iii) *Theory Three: The Internet will shift the power to Middlemen:* The Internet market makers (IMMs) will become the market gatekeepers. They exert downward pressure on prices in the market and control a large segment of the market. Traditional organisations will have no choice but to participate in this. These brand new organisations act as information and transaction brokers for a specific industry niche exploring information gaps.

In any case where the above is true or not, the Internet is becoming an elemental conduit that will tie an increasing number of enterprises together and accelerate the creation of virtual supply chains. No doubt, it will change the playing field.

How will successful enterprises form Internet-enabled supply chain strategies? Internet and application integration strategies will be driven by an enterprise's position with a community of common interest.

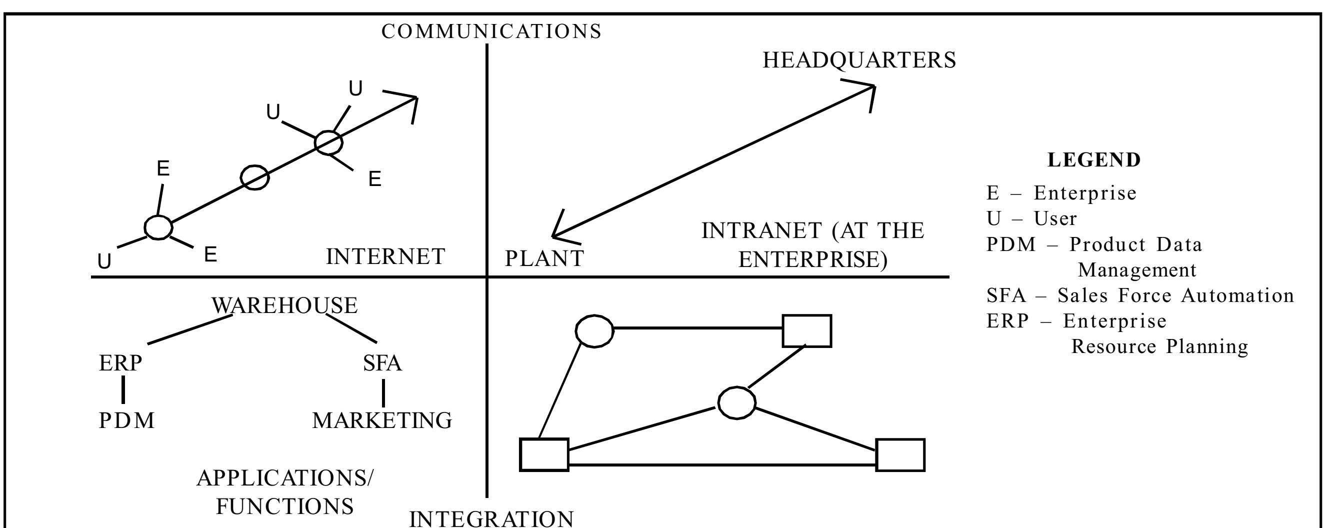

Fig. 12.17 – Supply Chain Foundation

Most enterprises will need to adopt different Internet behaviours depending on their position in the demand and supply spectrum. Enterprises will need to know:

- Examine each supply chain they participate in
- Analyse how the Internet may change the relative positions of supply chain participants
- Create and execute Internet-enabled supply chain strategies to take advantages of these changes.

12.16 TRANSFORMING SYPPLY CHAIN INTO AN INTELLIGENT, DIGITAL ECOSYSTEM

Start a Chain Reaction

Mr. Brian Houck opines on this subject that improving customer experience was important to client. Integrated decision analysis helped us identify the highest value decisions around the supply transformation, including the impact on customers, both as on date and for the future. Optimising supply chain to meet modern customer demands takes more than selecting technology and functional excellence. It requires transformation and integration across all of the operations. PWC's connected supply chain solution helps one to design and inplement a digital, intelligent, fully transparent, end-to-end and interconnected supply chain that optimises for speed, risk and profit.

A new dynamic ecosystem that uses advanced analytics to be more predictive, resilient and responsive, optimise for speed, risk and profit.

Unchain Operations

Siloed function, lack of partner visibility, products that show up in the wrong place at the wrong time – the result? Inefficient operations and dissatisfied customers. A traditional linear supply chain creates blind spots, gaps of knowledge and lack of real-time information. Bring down those walls and create a dynamic, adaptive and connected supply chain ecosystem with built-in intelligence and automation.

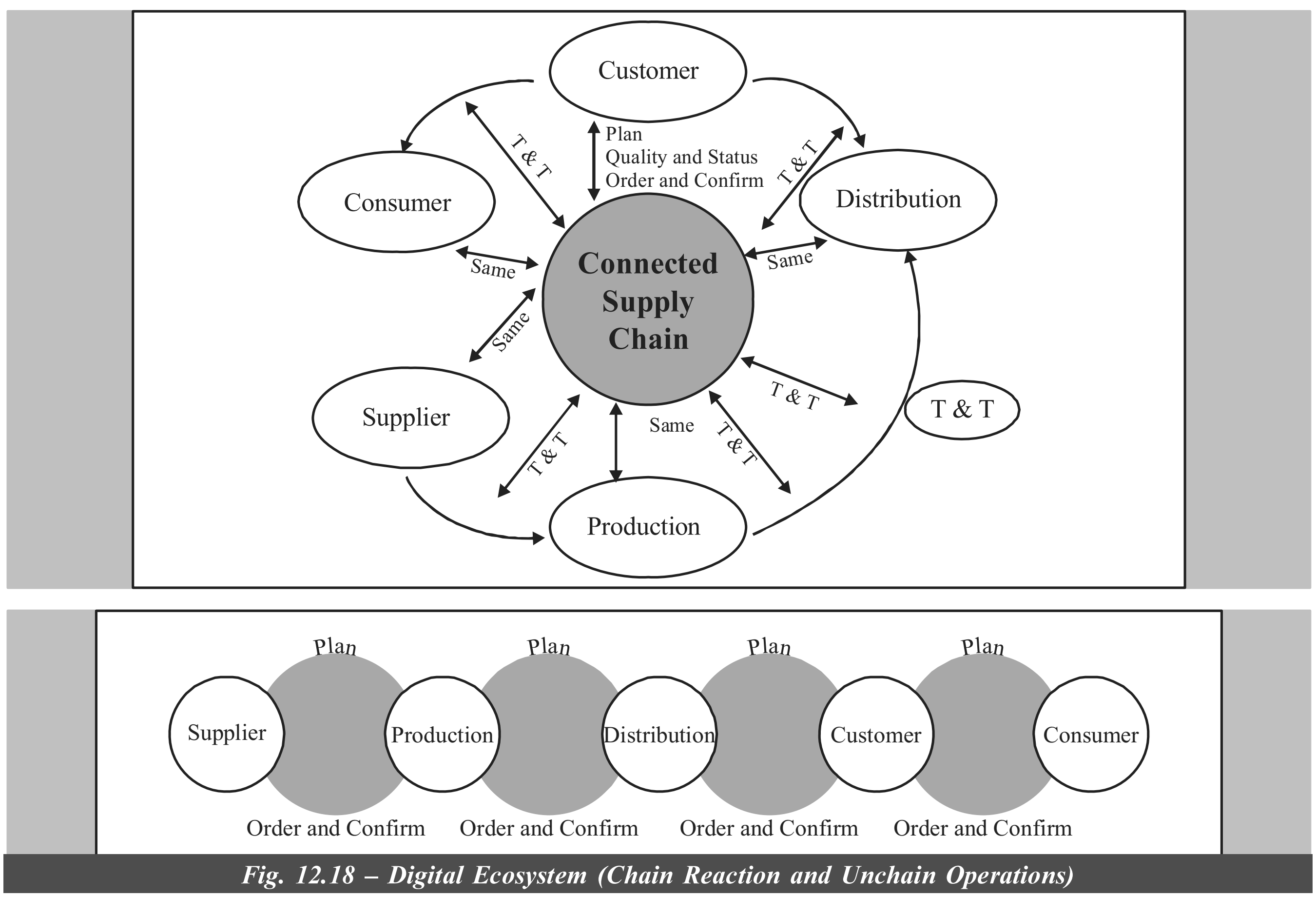

Fig. 12.18 – Digital Ecosystem (Chain Reaction and Unchain Operations)

THE LEAST YOU NEED TO KNOW

- *What is an Internet? How and when it took birth?*
- *What is supply chain? What are the different categories of supply chain?*
- *What is SCM? What are the goals, functions and benefits of SCM? Characteristics of SCM?*
- *An understanding of strategies of SCM, constraints programming and supply chain optimisation technology?*
- *What is an Integrated value chain?*
- *How supply chain is the precursor to integrated value chain?*
- *What is E-logistics? Its implementation in business houses?*
- *Understanding of SCP, SCE, SCIV strategic planning tools, tactical and execution planning tools, CTP, DTP, ATP, SCE Framework, SCM disconnects and supply chain portal.*
- *Study the SCM Systems' involvement in various activities and also the benefits, types, importance, significance, goals, objectives and roles.*

WRAP UP

- Supply chain management is the integration of the internal and external partners on the supply and process chain to get orders from customers, raw materials from suppliers and finished products to end-users. Supply chain management makes it possible for participants in the supply chain to see where the production, supply and inventory are in the process.
- Typical reasons for purchasing and implementing systems include:

Function	**Improvements Expected**
Inventory management	Cut inventory volume
Manufacturing management	(i) Ensure that products are delivered online in most efficient manner
	(ii) Cut manufacturing times
Procurement management	Increase revenues
	Reduction in costs of goods purchased
Distribution management	Improved sales and delivery time frames

- The reasons for supply chain management systems are:
  - Consumers and business demand improved service, more choice, and lower costs.
  - More parts and finished goods may have to be supplied to and from different locations around the world.
  - Internet and related technologies have a potential huge impact on the rate of change in distribution supply chains and consumer behaviour.
- SCM enables companies to avoid the vagaries of the marketplace by improved stock management techniques and better distribution management. This can be achieved by cooperative relationships that stem from inter-firm linkages with upstream and downstream suppliers and consumers to create very positive win-win situations.
- Effective management of inter-firm linkages in the supply chain can add value in numerous ways:
  - Lowering cost
  - Improved support
  - Improved quality
  - Stronger competitive positioning
  - Fast delivery
- How intranet-based e-commerce technology and supply chain functions are becoming intertwined, as the walls between functional structures (purchasing, personnel and production) are taken away.
- The dominant players in the SCM software in the horizontal and vertical marketplaces are SAP, Oracle, PeopleSoft and Baan (a Dutch firm). Application software, can of course, be deployed on a variety of platforms but, typically

in the current environment, it is deployed on a variety of mid-rays platforms manufactured by many different hardware vendors including DEC, Hewlett Packard and IBM.

- Supply chain planning tools are of three categories: Strategic planning tools, Tactical planning tools and Execution tools. Out of these three, today's vendor focus/hyper are on tactical planning tools. Enterprises have used strategic planning tools periodically as stand-alone solutions.
- Capable to promise systems enable on enterprise to commit orders against available material and capacity in real time. In this chapter, SCE Framework has been explained. SCP tells enterprises what to plan to do to meet business objectives under actual conditions. SCE tells enterprises what to do now and how to do it under actual conditions. Gartner Group has made an extensive study regarding SCM integration strategies and how the changes take place in the year 2005.
- Supply Chain Management involvement in E-commerce is also discussed. List of chain companies in 2018 are indicated.

TAKING STOCK

1. Define supply chain. List down the categories of supply chain and explain.
2. Define supply chain management? What are its characteristics? What are the benefits of supply chain management?
3. What are the goals, functions and strategies of supply chain management? Explain all the three in brief.
4. What are the benefits of using the Internet for SCM? Explain supply chain optimisation technology.
5. What is constraints programming? Explain the integrated value chain.
6. Why the supply chain is a precursor to Integrated value chain?
7. Explain E-logistics and its implementation in business houses?
8. 'SCM is a philosophy for improving the enterprises competitiveness.' Give your views.
9. 'Competition is building up in tomorrow's markets – may be a clash between competing supply chain.' Explain.
10. What is the integrated value chain? How does it differ from the supply chain?
11. In India, we have the IT initiative, the infrastructure initiative and the telecom liberalisation from the government. Do these contribute in any way to India's competitiveness in the international arena?
12. What does supply chain planning tools involve? Distinguish the three groups and their activities.
13. Define SCP and SCE. Compare and contrast the two.
14. What do you mean by CTP, ATP and ATO? Explain each one of them.
15. Explain collaborative planning initiatives.
16. Write down a sketch for SCM disconnects. Explain supply chain inventory visibility (SCIV).
17. What are SCM applications? Explain SCM markets.
18. Explain Transportation evolving from transactions to SCE with a sketch?
19. What are the shortcomings of TMS, WMS and OMS?
20. Explain SCE framework with a sketch.
21. Explain supply chain management involvement with regards to benefits, types, importance.
22. Explain SCM involvement with regards to significance, goals, objectives, role.
23. Explain Supply Chain Management System.
24. Define CRM System and E-commerce ERM systems.
25. Explain E-commerce ERM System.
26. How do you transform supply chain into an intelligent digital ecosystem?